Beyond Word Error Rate: A Failure Taxonomy of Streaming Dictation ASR

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Abstract

Speech recognition (ASR) systems are usually compared by word error rate (WER) on recorded files, after punctuation, capital letters and number format are removed. In dictation, the audio is streamed live and the text goes into the user's message as written, so what matters is which mistakes the user has to fix. We streamed 106 scripted English dictations from one speaker (read in a quiet room, with a fan on, and whispered), 20 clips in each of eleven other language and dialect sets, and 400 recordings that join clips in two languages to eleven ASR systems. We then sorted every error by a fixed rule into four levels: recognition, output formatting, language handling and, for the models we run ourselves, audio dropped before the model. Systems with similar WER failed in different ways. Zoom Scribe and OpenAI's GPT-Live-Transcribe have almost the same WER on quiet English (7.4% and 7.3%), but 34.7% and 8.8% on whispered English: Zoom and both AssemblyAI models left out 26–28% of whispered words, and still 21–26% with their end-of-turn detection relaxed or off, while seven other systems left out at most 3%. Qwen3-ASR left out 32%, but only because of the speech detection in our own deployment; relaxed, it left out 6%. With 107 speakers from the EARS corpus, the same four systems again lost 7–31% of whispered words, as did our self-hosted Whisper (10%), while six others lost under 2%. With only English selected, Gemini, Deepgram and our self-hosted Whisper lost the non-English part of a two-language recording in 86–99% of recordings, against 33–46% with both languages selected, while GPT-Live-Transcribe, OpenAI's realtime Whisper and Meta lost it in at most 20%. Running Whisper behind the Silero voice activity detector, set to drop audio it judges to be silence, raised whispered WER from 10.1% to 51.4%. Most English results come from one read speaker. We release the code, the recordings, both references and every transcript.

1 Introduction

This paper studies one use of ASR systems, which we call *dictation*: the user speaks a message, an email, a note or an instruction to an AI assistant, and the text is placed into another app. The transcript is the final product. Every error stays in the message unless the user finds it and corrects it, and correcting recognition errors is a large part of the cost of entering text by voice [55; 45].

WER counts errors but does not say what kind they are. On a dictation of eleven sentences of similar length, a WER of 9% could mean one wrong word in every eleven, or one whole sentence missing from otherwise correct text. A single average can also hide large differences between groups of speakers [22] and between dialects [52].

Public benchmarks run complete audio files through each system and report WER on normalised text [41; 50]. Dictation differs in four ways that we test:

- the audio is streamed, so we stream it at the speed of speech and measure how long the final text takes after the audio ends;
- the text is used as written, so we score numbers as written and also report errors in punctuation and capital letters;
- people dictate in different conditions, so we record English in a quiet room, with a fan on and in a whisper;
- people dictate in other languages and mix languages, so we test 11 language and dialect sets covering 7 languages, and recordings that switch from one language to another.

We test eleven systems: two open models that we run ourselves and nine services run by their providers. Every transcript is lined up with the correct text word by word, and each error is sorted by a fixed rule into one of four groups, which we call *levels*: recognition, output formatting, language handling, and the audio path before the model (Section 3). The levels describe what the output shows, not a proven cause.

The main finding is that systems with similar WER fail in different ways, and that the differences are large. Zoom and GPT-Live have almost the same quiet-English WER (7.4% and 7.3%), but on whispered English Zoom leaves out 28% of words and GPT-Live 1%. With only English selected, some systems remove the other language of a two-language recording almost every time and others keep it. And the app around the model matters: running Whisper behind a voice activity detector that drops audio it judges to be silence raised its whispered WER from 10.1% to 51.4%.

The paper makes four contributions:

- **A failure taxonomy for streaming dictation** that groups error types by the level they show up at, with a stated rule for detecting each type (Section 3).
- **A per-level comparison of eleven systems** on in-house data and on two external corpora with many speakers, including two behaviours known for open models [38; 42; 5] that, to our knowledge, have not been measured for commercial streaming systems: a language setting that removes the other language for some systems and only guides the language for others, and output in the wrong writing system (Sections 5–8).
- **Two references for the same speech**: the script the speaker meant (the *script reference*) and a record of what they said (the *spoken reference*), which separate a system that hears every word from one that writes the intended message (Section 7.3).
- **A case study of the level before the model**: nine ways of running Whisper that differ in hosting, in how the audio is cut, in which voice activity detector (VAD) is used, and in whether the VAD drops audio before the model (Section 9).

We release the test client, the analysis code, the English recordings, both references and every system's transcripts.

2 Related Work

ASR benchmarks and error analysis. Public leaderboards test open models and commercial services on complete audio files from public datasets, and clean up the text first so that differences in formatting do not count as errors [50; 41]. Independent tests of commercial services stream audio at the speed of speech and report speed alongside accuracy [4]. WER depends on how the text is normalised and on the test data [51], and WER on one test set is a poor guide to WER on others [26]. Other measures look past WER: match error rate and word information lost [34], semantic distance between reference and transcript [21], scores that split hallucinations by kind [24], and rates of runs of inserted or deleted words per hour [32]. Error analysis has

shown which words are hard to recognise, such as rare words and words near disfluencies [16], and work on detecting and correcting ASR errors is reviewed by Errattahi et al. [13]. Error categories close to ours have been applied to Whisper on South Asian languages, where output in the wrong writing system caused most failures in three of four languages [5], and wrong-script output has been measured across many model–language pairs [42] and removed from WER by mapping both texts to one script [37]; these studies test open models. Some benchmarks score text closer to how it is written: ESB keeps casing and punctuation [14], LibriSpeech-PC scores punctuation and capital letters [33], and Earnings-21 reports accuracy on named entities [11]. Most of these report how many errors a system makes on complete files. We report what kinds of errors commercial systems make in a streaming dictation setting, and we score the text as the user receives it, without converting numbers to a common form.

Dictation and text entry. In a study of short messages on phones, speech entry was about three times faster than typing, with slightly more errors left in the final text [45]; correcting recognition errors is a large part of the cost of speech entry [55]. This is why we care about which errors a user can see and fix.

Many languages and dialects. FLEURS [8] and VoxPopuli [56] provide read speech and parliament speeches in many languages. Arabic dialects remain much harder for current systems than Modern Standard Arabic [52; 1]. Arabic dialects also have no standard spelling, which has led to spelling conventions [19] and to scoring against several correct texts [2]. Urdu ASR has been benchmarked on read and conversational speech [3]. Urdu is also often written in Latin letters (Roman Urdu), and converting code-mixed Roman Urdu back to Urdu script is a task of its own [58].

Switching languages. Code-switching, moving between languages between or within sentences [39], has long been hard for speech and language technology [57]. Existing corpora, such as SEAME [29] and ASCEND [28] for Mandarin and English, contain natural switches within a sentence, for a few language pairs. For Whisper, the language given in the prompt changes code-switched output, and a single language can lead it to write a translation instead of a transcript [38]. WER over a whole recording can hide errors on the switched words, which has led to scoring only those words [54]. We instead build recordings that switch between sentences, for many language pairs. These recordings are less natural, but they reduce the chance that a language was lost because it is hard on its own.

Whispered speech. Whispered speech has no voicing and is much harder for recognisers trained on normal speech [27]; work on whispered ASR adapts models to it, for example with synthetic pseudo-whisper training data [15]. Corpora with whispered readings include wTIMIT [27], CHAINS [9], AISHELL6-Whisper for Mandarin [25] and EARS [44], which records the same text read normally and whispered by 107 speakers; we use EARS to compare the systems across speakers.

Further related work, on latency, partial text, the written form and running Whisper, is in Appendix A.

3 A Failure Taxonomy

We group error types into four levels, by where in the pipeline the error shows up (Table 1).

- **Recognition**: words missed, misheard or invented.
- **Output formatting**: words recognised but written in a form the user did not intend, such as a number in words instead of digits, or text in the wrong writing system. *Inverse*

text normalisation (ITN) is the step that turns spoken forms such as "seven" into written forms such as "7".

- **Language handling**: one language of a two-language message dropped, written in Latin letters, or translated.
- **Before the model**: audio the app never sends to the system.

The levels describe what we observe in the output, not a proven cause. For the hosted services, a missing word could have been lost in the service's own speech detection rather than in its model, and a language could be lost because of when the service finalises text rather than how it handles languages (Section 8). We can separate the level before the model from recognition only for the models we run ourselves, Whisper and Qwen, whose audio path we control (Sections 6.2 and 9). Every result therefore describes a system together with our integration of it (Appendix B).

Each type has a stated rule that a program applies to the word-by-word comparison. Three rules rely on earlier judgements: which words are names was decided by a language model following a written rule; the spoken reference was corrected by one listener; and the rules for romanised and translated parts were checked by hand only on examples (Appendix I).

Level	Error type	Rule	Section
Recognition	Wrong, missing, extra word (substitution, deletion, insertion)	a word replaced by another, left out, or added, from the word-by-word comparison	6.1
	Near-miss	a wrong word of 4 or more letters that differs from the correct word by 1 or 2 letters	6.1
	Name error	a word of a name in the correct text is not in the transcript	6.3
	Missing stretch	5 or more words in a row are missing (a run of deletions, as in the omission rate of 32), noted by position (first, middle or last third)	6.2
	Cut-off ending	the last 3 or more words are missing (truncation)	6.2
	Failed / lost clip	the clip's WER is at least 50% / at least 90%	6.2, 6.4
	Hard clip	the clip's WER is above 30%	6.4
	Invented text (hallucination)	text written for a clip with no speech, or a short stock phrase written in place of a whole clip's speech	9
Output formatting	Number format (ITN)	a number in digits in the correct text comes out in words, or the other way round	7.1
	Formatting	a wrong capital letter or punctuation mark (formatted error rate, Table 3)	5
	Clean-up (disfluency removal)	fillers, repeats and self-corrections removed: the difference between WER against the spoken reference and against the script reference	7.3
	Wrong writing system	most of the letters in the text belong to a different writing system from the chosen language's	7.2
Language handling	Missing part	less than 20% of the words of one language's part of a two-language recording are heard	8
	Romanised part (transliteration)	a missing part is replaced by Latin-letter text that is not English	8
	Translated part	a missing part is replaced by English (at least 4 words, at least 15% of them common English function words such as "the" or "of")	8
Before the model	Dropped by the app's VAD (VAD gating)	audio the app never sent to the system	9

Table 1: The failure taxonomy: error types grouped by the level they show up at, with the field's usual term in brackets, the rule used to detect each, and the section that measures it. The 50% bar for a failed clip is where half of the message would have to be retyped; the 30% bar for a hard clip counts a clip that needs several corrections; the 20% bar keeps a part with a few misheard words from counting as missing.

4 How We Ran the Test

Systems. We test eleven systems that transcribe in real time. Two are open models that we run ourselves: Qwen3-ASR 1.7B [48] ("Qwen") and Whisper Large V3 [41] ("Whisper SH", self-hosted). Nine are run by their providers: Meta Muse Voice Transcribe ("Meta"); Gemini 3.5 Transcribe (Live API; "Gemini"); Deepgram Nova-3; AssemblyAI Universal-3.6 Pro and Universal-3.5 Pro ("AAI 3.6" and "AAI 3.5"); Zoom Scribe (Live API; "Zoom"); ElevenLabs Scribe v2 Realtime ("ElevenLabs"); OpenAI GPT-Live-Transcribe (Realtime API; "GPT-Live"); and OpenAI GPT-Realtime-Whisper, OpenAI's hosted streaming Whisper ("RT Whisper"). Section 9 compares nine ways of running Whisper; the main comparison uses two of them, Whisper SH and RT Whisper. The main runs took place between 1 and 21 October 2026, the EARS and VOiCES runs and further Whisper and Gemini tests on 21–24 October, follow-up runs on 29 October, and on 5 November Qwen's two-language runs and the checks of GPT-Live's delay setting, of end-of-turn settings and of regional language codes (Appendix B).

We use each system's default settings, with a few changes given in Appendix B. Each system

is told the language a user would choose, as a language code without a region unless a system needs one; if a system has no option for that language, we send none and the system detects the language itself. Sending each dialect's own regional code, where a provider documents one, changed almost nothing (Appendix B).

Clean-up settings. Gemini and ElevenLabs write every word as spoken by default, including fillers, repeats and false starts, and each offers a setting that removes them: Gemini's Smart mode, which also adds punctuation, capital letters and structured formatting [17], and ElevenLabs' `no_verbatim` option [12]. We report these as two extra rows, "Gemini Smart" and "ElevenLabs NV" (no verbatim), so the result tables have thirteen rows. Where we count errors that all systems share, we count the eleven systems and leave these two rows out.

In-house English. 106 English messages, written by us to resemble chat, email, notes and prompts to AI assistants, were read aloud by one speaker, a member of our team, so the in-house English results describe one voice. Each message was recorded once, in one condition: 73 in a quiet room, 17 with a fan running nearby and 16 in a whisper. The three conditions use different messages, so we compare systems within a condition, not the conditions with each other; the fan-on set is a second set of read messages, not a test of noise. We call the third condition *whispered speech*; the model is always named by its setup (for example Whisper SH), to keep the two apart. The reader deliberately moved between US, South Asian, British and Irish English, often within a message (Appendix D).

Two references. Each English message has two correct texts. The *script reference* is the text the reader was given: what they meant to send. The *spoken reference* is what they actually said, made by correcting the script against the audio. The two differ in 85 of the 106 messages, by 453 of 10,776 words (4.2%): 242 added, 88 left out and 123 changed, including repeated words and spoken self-corrections. To make the spoken reference, a program proposed an edit wherever several systems heard the same departure from the script, and one listener accepted or rejected each proposal and corrected the rest by hand (Appendix C). Unless stated otherwise, English results use the script reference, because it is what a dictation user wants on the screen; Section 7.3 compares the two.

Other languages, two-language recordings and no-speech clips. We take 20 short clips from each of 11 language and dialect sets in public datasets (Table 2). From these we build 400 recordings that join two clips in different languages (Section 8). We also include 10 clips with no speech (silence and background noise) to check whether a system writes text when nobody is speaking. The single-language clips add up to 99 minutes of speech.

External corpora. Two public corpora with many speakers repeat the whispering and noise tests: EARS and VOiCES (Sections 6.2 and 6.5, Appendix F). We keep them apart from our own data and score them differently.

Set	Source	Clips
English, quiet	in-house	73
English, fan on	in-house	17
English, whispered	in-house	16
Arabic (Modern Standard, MSA; Egyptian speakers)	FLEURS [8]	20
Arabic (Saudi: 12 Najdi and 8 Hijazi clips)	SADA [1]	20
Arabic (Emirati)	Casablanca [52]	20
German, French, Hindi, Urdu	FLEURS	4×20
Mandarin Chinese	FLEURS	20
Spanish (Latin American)	FLEURS	20
Spanish (Spain)	VoxPopuli [56]	20
Spanish (Puerto Rico)	OpenSLR [18]	20
No speech	in-house	10
Two-language recordings	built from the clips above	400
External: whispered and normal	EARS [44]	2×428
External: clean, far, far + babble, far + TV	VOiCES [43]	4×100

Table 2: The test data. The 336 single-language clips contain 99 minutes of speech. The external corpora are kept apart from ours and scored differently.

Sending the audio. Each clip is streamed at the speed of speech in small chunks, the way a microphone delivers it, with a new session for every clip and all systems called from the same client machine (Appendix B). A system *finalises* text when it commits to it and stops revising it; we score only finalised text.

A voice activity detector (VAD) is a small model that decides whether audio contains speech. Except where Section 9 says otherwise, no VAD drops any audio: every system receives every sample. Whisper SH uses Silero VAD [49] only to decide where to cut the audio into segments. Section 9 measures what happens when a VAD also drops audio, using a second Whisper deployment of ours with a VAD in front of it on the client ("gated Whisper").

Scoring. Before comparing a transcript with the correct text, we ignore capital letters and punctuation, and we treat variant letter forms in Arabic, Urdu and Chinese as equal (Appendix C). We do not convert numbers: we score the text as written, so "seven" in place of "7" counts as an error.

Each transcript is lined up with the correct text (the reference) using the fewest possible edits (a Levenshtein alignment). This gives three kinds of word error: a *substitution* (a wrong word), a *deletion* (a missing word) and an *insertion* (an extra word). WER is the total number of errors in a set divided by the total number of words in its correct text. It can exceed 100%, because extra words also count as errors. Mandarin, which is not written with spaces, is scored by counting each Chinese character as one unit (Appendix C). The *character error rate* (CER) is the same measure counted over characters. We also report a *formatted* error rate, a case- and punctuation-sensitive WER as in ESB [14]: words keep their capital letters and each punctuation mark counts as a separate token, so a wrong capital or a missing comma each counts as one error.

Uncertainty. For English WER we give a 95% confidence interval from a percentile bootstrap over clips with 2,000 resamples [7] (4,000 for the significance tests). These intervals describe variation across messages, not across speakers. When we compare two systems, or one system under two conditions, we resample the same clips for both (a paired comparison). A difference is *clear* when the 95% interval of the paired difference excludes zero and it still holds after a Holm correction for the number of comparisons in its group (for example, the 13 name gaps form one

group; Appendix C); otherwise it is *within chance*. The language sets have only 20 clips each, so differences of a few points there are within chance. Differences are given in percentage points ("points").

5 One Score, Different Failures

Similar WER on quiet English, different WER elsewhere. Table 3 gives each system's WER and formatted error rate for each English condition. On quiet English, ten of the thirteen rows are between 3.9% and 9.5%, and the order of these systems tells little about the other conditions. Zoom (7.4%) and GPT-Live (7.3%) are the clearest case: on whispered English, Zoom's WER is 34.7% and GPT-Live's 8.8% (paired difference 25.8 points, 95% CI 10.3 to 51.7), almost all of it missing words. AAI 3.6 (8.2%) and RT Whisper (8.2%) are another: 30.5% and 8.9% whispered. Gemini Smart has the lowest quiet-English WER (3.9%), clearly below the next row, ElevenLabs NV; the lowest with default settings is Gemini, at 6.9%. When capital letters and punctuation are also scored, error rates on quiet English rise to 8.8–22.8%; Gemini Smart (8.8%), ElevenLabs NV (9.0%) and Gemini (9.1%) are lowest.

	WER			Formatted		
System	Quiet	Fan on	Whispered	Quiet	Fan on	Whispered
Whisper SH	8.3 [6.8, 9.8]	6.5 [4.1, 10.0]	10.1 [7.1, 13.0]	16.0	14.3	20.8
Qwen	13.6 [10.9, 17.0]	7.0 [5.3, 9.1]	46.6 [32.9, 67.5]	21.6	15.5	56.5
Meta	9.5 [8.3, 10.6]	7.4 [5.3, 9.6]	11.3 [9.0, 14.1]	18.3	14.9	22.4
Gemini	6.9 [5.8, 8.0]	2.5 [1.8, 3.3]	5.0 [2.8, 6.7]	9.1	6.4	8.6
Gemini Smart	**3.9** [3.2, 4.7]	3.2 [2.1, 4.6]	4.2 [2.9, 5.7]	8.8	10.0	10.9
Deepgram	13.6 [12.0, 15.1]	11.4 [8.8, 14.0]	19.8 [15.7, 23.6]	22.8	20.0	32.1
AAI 3.6	8.2 [7.2, 9.1]	4.1 [2.7, 5.7]	30.5 [14.6, 55.8]	13.1	8.8	35.1
AAI 3.5	9.0 [7.9, 10.1]	4.3 [2.8, 6.2]	30.9 [15.2, 55.9]	14.9	9.4	35.9
Zoom	7.4 [6.5, 8.3]	3.6 [2.2, 5.2]	34.7 [19.2, 60.2]	13.7	10.2	41.6
ElevenLabs	12.8 [10.9, 14.9]	5.8 [3.7, 8.2]	9.1 [5.4, 12.9]	21.9	11.6	16.0
ElevenLabs NV	5.1 [4.3, 5.9]	4.5 [3.1, 6.0]	7.1 [3.6, 10.7]	9.0	8.9	12.4
GPT-Live	7.3 [6.3, 8.3]	3.6 [2.4, 4.8]	8.8 [7.0, 10.4]	11.2	8.1	14.2
RT Whisper	8.2 [7.0, 9.5]	4.6 [3.2, 6.4]	8.9 [7.2, 11.0]	11.9	8.1	13.8

Table 3: English error rates (%), lower is better. WER is shown with its 95% confidence interval. Bold: clearly below the next-lowest system (paired bootstrap); this holds only for quiet-English WER. Formatted: WER that also counts mistakes in capital letters and punctuation. Whisper SH: our self-hosted Whisper Large V3; RT Whisper: OpenAI GPT-Realtime-Whisper.

Other languages. For European languages and Mandarin the systems are close, and most stay below 10%. Hindi ranges from 2.6% to 34.6%, Urdu is above 85% for several systems (Section 7.2), and the Saudi and Emirati Arabic sets are hard for every system (Section 6.4). No system is clearly best on any language set. Appendix E gives WER for each set (Table 15).

Speed. Measured from our test client, typical (median) waits from the end of the audio to the final text range from 313 to 760 ms. Meta, Deepgram, Whisper SH, Qwen and both ElevenLabs settings wait under 400 ms; GPT-Live and RT Whisper about 750 ms. Appendix E.1 gives the full timings, which we treat as indicative.

6 Recognition

6.1 Wrong, Missing and Extra Words

WER adds three kinds of error together: wrong words (substitutions), missing words (deletions) and extra words (insertions). Table 4 splits each system's WER into these three parts for quiet English, English with the fan on, and the other languages.

	Quiet English			English, fan on			Other languages		
System	Wrong	Missing	Extra	Wrong	Missing	Extra	Wrong	Missing	Extra
Whisper SH	3.2	2.2	2.9	4.2	0.8	1.5	12.1	3.7	0.9
Qwen	5.0	2.7	6.0	4.6	0.4	2.0	11.7	2.5	0.9
Meta	3.5	1.1	4.9	4.6	0.6	2.2	9.4	2.6	1.0
Gemini	2.1	1.1	3.7	1.6	0.5	0.5	7.8	2.0	0.9
Gemini Smart	1.9	1.3	0.7	1.8	1.1	0.3	7.6	3.2	0.6
Deepgram	5.8	2.5	5.3	7.0	1.8	2.6	9.2	5.7	0.9
AAI 3.6	2.8	1.3	4.0	2.5	0.6	1.0	9.4	1.9	0.6
AAI 3.5	2.9	1.4	4.7	2.5	0.8	1.1	12.1	2.2	0.7
Zoom	2.6	1.4	3.4	2.5	0.5	0.7	10.3	2.8	1.0
ElevenLabs	2.8	1.9	8.1	3.5	0.5	1.9	11.1	1.9	1.2
ElevenLabs NV	2.5	1.2	1.4	3.0	0.5	1.0	13.1	4.0	1.2
GPT-Live	2.9	1.2	3.2	2.5	0.5	0.6	8.6	3.6	0.6
RT Whisper	2.9	1.2	4.1	3.0	0.8	0.8	9.1	3.2	0.5

Table 4: Wrong, missing and extra words (substitutions, deletions, insertions) against the script reference, each as a share (%) of all words in the correct text. For each English condition, the three add up to the WER, give or take 0.1 for rounding. Other languages: the 10 non-English sets other than Urdu, pooled by words, so the sum differs slightly from the per-set average in Table 15 (Urdu errors are mostly the wrong writing system; Section 7.2).

Quiet English. Against the script reference, extra words are the largest part of the error for nine of the thirteen rows: Gemini, for example, has 3.7% extra words against 2.1% wrong ones, and ElevenLabs 8.1% against 2.8%. Many of these extra words were really said: they are repeats and self-corrections the reader added. Against the spoken reference, extra words are the largest part for only four rows (Gemini, AAI 3.5, Zoom and ElevenLabs): Gemini's extra words fall from 3.7% to 1.6%, but ElevenLabs' only to 5.7%, so most of ElevenLabs' extra words were not said. The mix also differs between systems with the same WER: Whisper SH and AAI 3.6 (8.3% and 8.2%) differ by only 0.1 points in WER, but Whisper SH misses more words (by 0.9 points, 95% CI 0.1 to 2.4) and adds fewer (by 1.1 points, 95% CI 0.5 to 1.8). Missing words are between 1.1% and 2.7% for every system.

With a fan on and in other languages. On the fan-on set, wrong words are the largest part for all thirteen rows; Deepgram has the most, 7.0%. Outside English, wrong words are also the largest part for every system: 7.6–13.1% of words, against 1.9–5.7% missing and 0.5–1.2% extra.

What the wrong words are. For every system, ordinary words are the largest group of wrong English words (46–63%). Numbers written as words instead of digits or the other way round (Section 7.1), and near-misses such as "request" for "requests", together make up 33–50%; names are 3–5% (Appendix E, Table 19).

6.2 Whispered Speech

Four systems leave out whispered words. On whispered speech, Qwen, Zoom and both AAI models leave out 26–32% of words; every other row leaves out at most 3% (Figure 1). With 16 clips these shares are uncertain (for Qwen, 95% CI 15–58%), but each of the four intervals lies above 3% (this also holds with bias-corrected intervals and after a Holm correction). Their errors are concentrated in some clips: 5 or 6 of the 16 whispered clips fail (WER of 50% or more), and 2–4 are almost entirely lost (90% or more). In the other group, no whispered clip fails, and wrong words are the largest part of the error for every row except Gemini Smart, whose few errors are split about evenly between wrong and missing words.

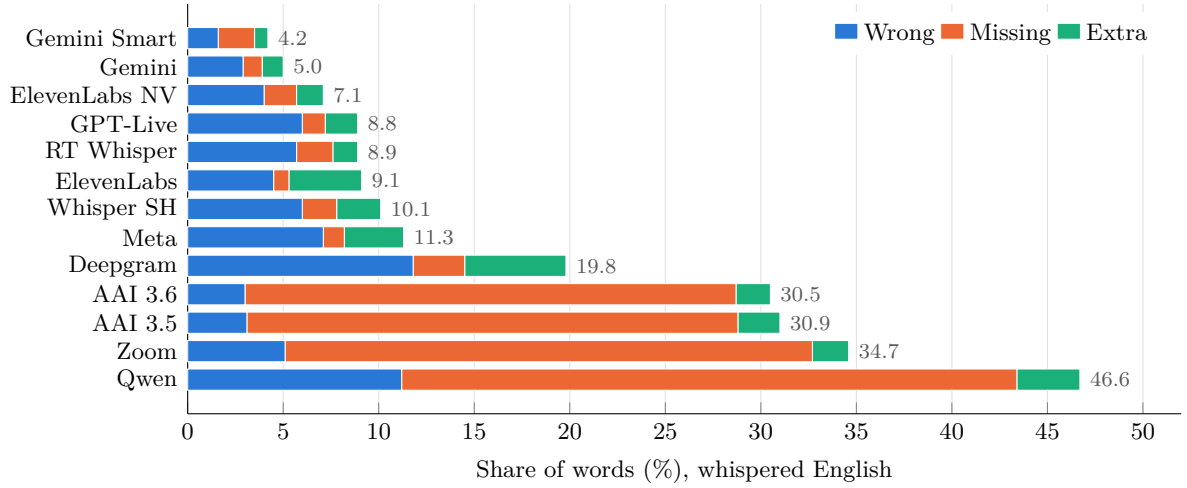


Figure 1: Whispered WER split into wrong, missing and extra words, as a share of all words; the number is the WER. Four systems lose most of their whispered words as missing words; the rest mostly get words wrong.

Where the words go missing. Both AAI models cut off the last 3 or more words of 4 of the 16 whispered clips. Qwen's missing stretches appear anywhere, including at the start. A cut-off ending leaves no visible gap in the text, and WER does not show it: AAI 3.6 (30.5%) and Zoom (34.7%) have similar whispered WER, but 4 of AAI 3.6's whispered clips have a cut-off ending and none of Zoom's (too few clips to test).

Recognition, turn detection or speech detection? Qwen, Zoom and the AAI models all decide on the server when speech or a turn has ended, so a missing whispered word could be lost there rather than in recognition. On 5 November we tested this on the 16 whispered clips (Appendix B). For AAI, forcing the end of the last turn before closing the session changed nothing (AAI 3.6: 25.7% of words missing, then 25.6%), and turning end-of-turn detection off (`end_of_turn_confidence_threshold` 1, 10 s maximum silence) left 22.7% missing for AAI 3.6 and 25.7% for AAI 3.5. For Zoom, ending a segment after 5 s of silence instead of 600 ms lowered missing words from 27.6% to 21.2%. So AAI and, for the most part, Zoom lose these words whatever their turn settings; since we cannot see inside a hosted service, we count this as recognition, which here includes any speech detection inside the service. Qwen is different: our deployment runs its own speech detection before the model, with default settings, and relaxing it (threshold 0.3, 2 s of silence) lowered Qwen's missing words from 32.2% to 5.7%. Qwen's whispered loss is therefore in the audio path of our own deployment, the level before the model, as for gated Whisper (Section 9). The relaxed setting cost nothing we could measure: on the 46 quiet and 14 fan-on messages that completed (the longest timed out on our single server), WER was 10.3% against 11.3% and 5.6% against 6.3% with the default, and it wrote no text on the 10 no-speech clips. The tables keep the deployment's default setting.

The same pattern across 107 speakers. EARS [44] has 107 speakers recorded in a soundproofed studio, reading the same passage (the Rainbow Passage) normally and in a whisper; we use parts 1–4 from every speaker, 428 normal and 428 whispered clips. For EARS we normalise numbers and spellings in reference and transcript with Whisper's English text normaliser and report only WER; intervals resample speakers. Table 5 gives the results. Normal speech is easy for every system here (0.8–2.5%). Every speaker reads the same well-known passage, which a system may have seen in training; we cannot check this, so EARS tests the same text across many voices, not unseen text. Whispered, the same four systems again lose words, now 7–31% of them: AAI 3.6 and AAI 3.5 rise by about 7 points, Zoom by 24 and Qwen by 31 (with its deployment's default speech detection, as in the tables). Whisper SH, which lost few words on our whispered clips, loses 10.4% of words as missing on EARS, against 7.3% for each AAI model (within chance), so the two groups of our in-house test are not a clean split across speakers. Gemini (in both settings), ElevenLabs (in both settings), Meta, GPT-Live and RT Whisper change by less than one point, and no speaker reaches a whispered WER of 20% with any of them.

System	Normal	Whispered	Change (paired)	Whispered: wrong / missing / extra	Speakers ≥20%
Whisper SH	1.5	11.5 [8.2, 15.4]	+10.0 [+6.5, +14.0]	0.8 / 10.4 / 0.3	20
Gated Whisper	1.0	42.9 [36.7, 49.4]	+41.9 [+35.1, +48.4]	3.1 / 39.1 / 0.7	67
Qwen	1.1	32.0 [26.3, 38.1]	+30.9 [+25.0, +37.0]	1.0 / 30.6 / 0.4	56
Meta	1.4	1.7 [1.3, 2.1]	+0.3 [0.0, +0.6]	0.8 / 0.3 / 0.5	0
Gemini	0.9	1.1 [0.9, 1.4]	+0.2 [+0.0, +0.5]	0.5 / 0.2 / 0.3	0
Gemini Smart	0.8	1.0 [0.8, 1.3]	+0.2 [0.0, +0.5]	0.6 / 0.2 / 0.2	0
Deepgram	2.5	4.0 [3.2, 4.8]	+1.5 [+0.8, +2.4]	1.8 / 1.5 / 0.6	1
AAI 3.6	1.2	8.1 [5.2, 11.3]	+6.9 [+4.1, +10.2]	0.5 / 7.3 / 0.3	17
AAI 3.5	1.2	8.3 [5.6, 11.6]	+7.1 [+4.3, +10.4]	0.6 / 7.3 / 0.4	16
Zoom	0.9	25.0 [20.1, 30.6]	+24.1 [+19.0, +29.9]	1.3 / 23.3 / 0.4	45
ElevenLabs	1.3	1.5 [1.1, 1.8]	+0.1 [−0.2, +0.5]	0.7 / 0.2 / 0.5	0
ElevenLabs NV	1.1	1.2 [0.9, 1.6]	+0.2 [−0.2, +0.5]	0.7 / 0.2 / 0.3	0
GPT-Live	1.3	2.1 [1.7, 2.5]	+0.8 [+0.4, +1.2]	1.2 / 0.4 / 0.4	0
RT Whisper	1.2	2.1 [1.7, 2.5]	+0.9 [+0.5, +1.3]	1.2 / 0.4 / 0.4	0

Table 5: EARS: the same 107 speakers reading the same passage normally and whispered. WER (%), numbers and spellings normalised; 95% intervals from resampling speakers. Change: whispered minus normal, in points, paired by speaker and passage. Speakers ≥20%: speakers whose whispered WER is 20% or more. Gated Whisper: gated Whisper with its VAD dropping audio (Section 9).

6.3 Names

Names are written correctly less often than other words. Our English scripts name people, products and apps (for example, "send it to Maria" or "post it in Notion"). Over all English dictation, every system writes names correctly less often than other words (Table 6). Deepgram, used without its custom-vocabulary option, has the largest gap: 9.7 points (95% CI 3.3 to 16.7). Five more systems have gaps of 4.8–6.5 points with intervals that exclude zero, but these are 13 comparisons, and after a Holm correction only Deepgram's gap remains clear. Names are 183 of 10,776 words, so even Deepgram's gap adds only about 0.2 points to its WER: a system can write many names wrongly and still have a low WER. Much of the gap comes from fan-on and whispered messages: on quiet English alone (123 name words), no system's gap is clearly above zero (Deepgram's is the largest, 6.4 points, 95% CI −0.4 to 14.7), and Whisper SH's and Qwen's fall from 6.5 to under 2 points.

Finding the names. We took every word in the English scripts that starts with a capital letter in the middle of a sentence: 327 words, 184 different ones. Each different word was

labelled as a name or not by a written rule (Appendix C); only 156 of the 327 are names, so the capital-letter rule alone finds names poorly. The 183 name words we score are these 156, the 22 occurrences of a listed name at the start of a sentence or after a title such as "Dr.", and 5 lower-case uses of a word that names a product in the same message (such as "app"), which our counting also includes.

System	Names	Other words	Gap	95% CI
Whisper SH	88.0	94.5	6.5	[1.1, 12.9]
Qwen	82.0	88.4	6.5	[0.1, 14.1]
Meta	90.2	95.0	4.8	[0.3, 10.2]
Gemini	93.4	96.9	3.5	[−0.3, 8.2]
Gemini Smart	95.1	96.8	1.7	[−1.4, 5.6]
Deepgram	81.4	91.1	9.7	[3.3, 16.7]
AAI 3.6	92.3	93.0	0.7	[−4.2, 6.0]
AAI 3.5	92.3	92.9	0.6	[−4.2, 6.2]
Zoom	90.7	92.7	2.0	[−3.3, 8.0]
ElevenLabs	91.8	95.4	3.6	[−0.4, 8.5]
ElevenLabs NV	94.5	96.1	1.6	[−1.8, 5.6]
GPT-Live	90.2	95.7	5.6	[1.5, 10.8]
RT Whisper	90.7	95.6	4.9	[0.6, 10.3]

Table 6: Share of words written correctly (%) for the 183 name words and for all other words in English dictation, and the gap in percentage points, with its 95% confidence interval (bootstrap over the 106 messages).

6.4 Dialects and Errors Every System Shares

Most failed clips are in four sets. In the eleven other language sets, WER is below 5% on several European sets and near or above 50% on Emirati Arabic and, for some systems, Urdu. Almost all failed clips are in four sets: Saudi Arabic, Emirati Arabic, Hindi and Urdu (Appendix E, Table 18). Urdu and Hindi are about writing (Section 7.2); the Arabic dialects are about recognition.

Arabic dialects. Saudi and Emirati Arabic have failed clips for every one of the eleven systems: 15–35% and 50–75% of clips. On Emirati Arabic, WER is between 48.8% and 61.8% with default settings. Part of the error may come from the correct text itself: Arabic dialects have no standard spelling [19], so a correct transcript can be spelled differently from ours. Each dialect also comes from a single corpus, so we cannot separate the dialect from how its corpus was recorded.

Errors every system makes. If every system fails on a clip, switching systems will not help; if only one fails, another system avoids it. Of the 220 non-English clips, 21 are hard clips (WER above 30%) for all eleven systems: 14 Emirati Arabic, 6 Saudi Arabic and 1 Spanish (Spain). Across all 326 speech clips, 23 are hard for exactly one system (Appendix E). In English, 67 of 73 quiet clips and all 17 fan-on clips are below 30% for every system, while 13 of 16 whispered clips are above it for at least one: for English, the choice of system matters most on whispered speech.

6.5 Distance and Noise

A far microphone, babble and a television. VOiCES [43] replays audiobook sentences in a room and records them with a far microphone, in quiet, with several people talking (babble) and with a television playing. With babble, WER rises for every system, to between 38.7% and

94.9%, mostly as missing words; Zoom is lowest with babble and with television, although it is worse than average with a far microphone in a quiet room. Gemini returned no text within 30 seconds on 67 of the 100 babble clips, which count as empty transcripts; on the other 33 its WER is 40.3%, so most of its babble WER is a failure to answer rather than misrecognition. Appendix F gives the full results.

7 Output Formatting

7.1 Number Format

Each system has its own habit. Most benchmarks convert numbers to a common form before scoring; we do not, so we can see how each system's ITN writes them. The English scripts contain 276 numbers written in digits and 193 written as words (such as "first" or "twenty"). Qwen writes 68% of the digit numbers as words ("seven" for "7"), Meta 47% and GPT-Live 45%; the two AAI models do the opposite, turning 41–42% of number words into digits; Whisper SH, Gemini, Zoom and ElevenLabs usually keep the form in the script (Appendix E, Table 20). These switches make up 16.1% of GPT-Live's English errors, 12.9% of Meta's and 10.4% of Qwen's, against 4.8–6.1% for ElevenLabs, Zoom and Whisper SH; a number written in several words counts once, so these shares are a minimum. The number itself was heard correctly.

The order of systems barely changes. Re-scored after Whisper's English text normaliser, which writes numbers as digits and removes fillers, every quiet-English WER falls, by 1.0–3.8 points when both scores use the same word splitting, but the order of systems changes little (rank correlation 0.98; Appendix E, Table 17). Number habits matter to a user who sees the text, but they are not what separates the systems on quiet English.

7.2 Writing System

Urdu. Urdu shows the biggest gap between systems, and the cause is mostly the writing system. Table 7 counts, for each system that writes Urdu in another script, the clips written mostly in Devanagari, the script of Hindi. Qwen, Meta and AAI 3.5 write most Urdu clips in Devanagari and fail on 85–95% of them. Meta and AAI 3.5 have no Urdu option and detect the language themselves; Qwen does this even when Urdu is selected. Spoken Urdu and Hindi are close, so a system detecting the language itself may reasonably hear Hindi; but a user who chose Urdu receives text in the wrong writing system, whatever words it contains. Zoom, which also has no Urdu option, writes 19 of 20 Urdu clips in Latin letters ("Ek bum governoral journal ke daftar…") and fails on every one. Both Whisper rows, Deepgram, AAI 3.6 and ElevenLabs write in the Urdu writing system and fail on at most 5% of clips; RT Whisper has the lowest Urdu WER (12.3%).

System	Urdu option	Mostly Devanagari	Failed (WER ≥50%)
Qwen	selected	17	19
AAI 3.5	none	18	18
Meta	none	17	17
Gemini	none	9	9
Gemini Smart	none	4	5
GPT-Live	selected	1	1

Table 7: Urdu clips (of 20) written mostly in Devanagari, and Urdu clips with a WER of 50% or more. Gemini Smart writes some Devanagari on a fifth clip, and GPT-Live on 3 clips in all. Zoom writes 19 of 20 in Latin letters. Source: `paper/analysis/trace.py`.

Hindi. Zoom has the highest Hindi WER of the eleven systems (34.6%), with text in the right writing system. Its CER (16.3%) is less than half its WER, so many of its wrong words differ from the correct word by a few characters, for example in vowel signs, rather than being different words (Appendix E).

7.3 Clean-up and the Two References

The reference decides which system looks best. The script reference asks whether the text is the message the user meant; the spoken reference asks whether the system heard what was said. A clean-up setting removes fillers, repeats and self-corrections on purpose, so it scores well on the first and loses points on the second. In the literature this is disfluency removal, and the choice between verbatim and clean transcripts [20].

Against the script. Smart lowers Gemini's quiet-English WER from 6.9% to 3.9%, and `no_verbatim` lowers ElevenLabs' from 12.8% to 5.1%, in both cases almost all through fewer extra words.

Against what was said. Table 8 scores the same transcripts against the spoken reference. Every system run at its verbatim default does better on quiet English, by 2.2–4.0 points, because words the reader added are no longer counted as extra. Gemini Smart is the only row that does worse (+0.8 points, paired 95% CI −0.1 to 1.6): it leaves out the repeats and self-corrections that are now in the reference. On quiet English, Gemini (3.7%), ElevenLabs NV (4.5%) and Gemini Smart (4.6%) are now close. In paired comparisons, Gemini is 0.9 points below Gemini Smart (95% CI −1.8 to 0.0), and Gemini Smart and ElevenLabs NV are 0.1 points apart. Against the script, Gemini Smart is clearly below ElevenLabs NV (by 1.2 points, 95% CI 0.6 to 1.9); against what was said, no clean-up setting is clearly ahead. The whispered results of Section 6.2 hold against either reference: every whispered WER moves by at most 1.5 points.

How much the spoken reference depends on the systems. The spoken reference was drafted from where the verbatim systems agreed, so it could favour them. Each system voted for 237 to 313 of the 378 accepted edits. We rebuilt the reference eleven times, each time keeping only the accepted edits that the other ten systems would still have proposed: no system's WER against its own rebuilt reference rose by more than 0.3 points in any condition (1.2 points for Meta on quiet English with the stricter thresholds used to pre-tick proposals). But the order of Gemini, Gemini Smart and ElevenLabs NV is not stable: without Gemini's votes, Gemini is 0.5 points below Gemini Smart (95% CI −1.3 to 0.3), and with the stricter pre-tick thresholds, it is 0.8 points above (95% CI −0.1 to 1.7); neither difference is clear. A clean-up setting is clearly best only against the script.

	Quiet		Fan on		Whispered	
System	Script	Spoken	Script	Spoken	Script	Spoken
Whisper SH	8.3	6.1 [4.9, 7.7]	6.5	4.6 [2.8, 6.8]	10.1	9.3 [6.2, 12.5]
Qwen	13.6	9.8 [7.2, 12.9]	7.0	5.1 [3.7, 6.6]	46.6	45.5 [31.6, 66.5]
Meta	9.5	5.5 [4.7, 6.4]	7.4	4.8 [3.1, 6.8]	11.3	9.9 [7.7, 12.0]
Gemini	6.9	3.7 [3.0, 4.4]	2.5	1.9 [0.8, 3.1]	5.0	4.6 [2.4, 6.1]
Gemini Smart	3.9	4.6 [3.8, 5.5]	3.2	3.3 [1.7, 5.3]	4.2	4.4 [2.9, 6.4]
Deepgram	13.6	9.6 [8.3, 10.8]	11.4	9.3 [6.8, 11.6]	19.8	18.3 [14.3, 22.0]
AAI 3.6	8.2	4.8 [4.0, 5.6]	4.1	1.7 [1.1, 2.2]	30.5	29.5 [13.1, 54.4]
AAI 3.5	9.0	5.6 [4.8, 6.4]	4.3	2.0 [1.3, 2.5]	30.9	29.9 [13.6, 55.7]
Zoom	7.4	4.8 [4.1, 5.6]	3.6	2.3 [1.1, 3.8]	34.7	33.5 [17.0, 59.6]
ElevenLabs	12.8	8.8 [7.1, 10.8]	5.8	3.3 [2.0, 4.8]	9.1	7.6 [4.1, 10.7]
ElevenLabs NV	5.1	4.5 [3.8, 5.4]	4.5	2.5 [1.4, 3.8]	7.1	6.6 [2.6, 10.8]
GPT-Live	7.3	5.0 [4.2, 5.9]	3.6	3.6 [2.1, 5.4]	8.8	8.1 [6.3, 9.5]
RT Whisper	8.2	5.0 [4.1, 5.9]	4.6	3.7 [1.9, 5.8]	8.9	8.4 [6.7, 10.4]

Table 8: English WER (%) against the script reference (as in Table 3) and against the spoken reference, with its 95% confidence interval (bootstrap over messages). Lower is better.

8 Language Handling: Two Languages in One Recording

With only English selected, some systems remove the other language. We call the English-only setting a *filter* for a system when, with only English selected, the other language is lost in more than half of the EN + X recordings, against fewer than half with both languages selected; and a *preference* when, with only English selected, it is lost in fewer than half. By this rule, the setting is a filter for Gemini (in both settings), Deepgram and Whisper SH: the non-English part is lost in 86–99% of the 100 recordings with only English selected, against 33–46% with both selected (Table 25, first column; Table 9 counts a loss of either part). It is a preference for Qwen, Meta, AAI 3.6, AAI 3.5, GPT-Live and RT Whisper, which lose it in 7–38%. Zoom and ElevenLabs (in both settings) lose most second languages whatever is selected, for reasons given below. Whether a filter is right depends on the app; a user who selects English and then says a sentence in another language may or may not want it written.

How the recordings were built. Our recordings are not natural code-switching [39; 57]: they join single-language recordings from different speakers, with a 300 ms pause between them, so the language, the speaker and the recording all change at a sentence boundary. 100 recordings pair English with one of the 10 other language sets (EN + X), and 300 pair two non-English languages (X + Y). Urdu is left out because several systems write it in the wrong writing system even alone (Section 7.2). English comes first in half of the EN + X recordings, and the order of each X + Y pair is balanced likewise. Each recording is played with both languages selected and with automatic detection, and EN + X recordings also with only English selected. All eleven systems were run on this test.

To score a recording, we line up the transcript with the two parts' correct texts joined end to end, and mark each reference word as heard, replaced or missing. A part is *missing* when less than 20% of its words are heard: either nothing is written in its place, or other text is. Table 9 gives the share of recordings in which a part goes missing. A part may also go missing because it is hard on its own; Appendix G repeats the analysis on the 85 EN + X and 232 X + Y recordings whose parts every one of the eleven systems gets at least 50% right when played alone (the *screened* recordings); the order of the systems is almost the same and the conclusions do not change.

	EN + X (100 recordings)			X + Y (300 recordings)	
System	Both selected	English only	Auto-detect	Both selected	Auto-detect
Whisper SH	45	87	49	87	88
Qwen	20	38	20	46	46
Meta	19	23	21	32	35
Gemini	50	93	71	36	75
Gemini Smart	52	87	73	62	83
Deepgram	40	99	40	58	58
AAI 3.6	10	25	15	28	28
AAI 3.5	13	35	16	23	33
Zoom	62	62	62	85	85
ElevenLabs	62	85	62	93	93
ElevenLabs NV	80	93	79	96	97
GPT-Live	**6**	**7**	**7**	**4**	**5**
RT Whisper	11	12	10	11	11

Table 9: Share of two-language recordings (%) in which one of the two languages went missing; lowest in each column in bold. EN + X: 100 recordings of English and another language; X + Y: 300 recordings of two other languages. For Deepgram, RT Whisper and Zoom, the both-languages and automatic settings send the same request, and Qwen wrote the same text in both (Appendix G). For shares between 25% and 75%, the 95% interval is about ±12–22 points for EN + X and ±8–11 points for X + Y; it is wide because recordings reuse source clips (each English message appears in 10 EN + X recordings), so the bootstrap resamples the source clips.

Which part goes missing. Table 25 (Appendix G) shows which part goes missing and what appears in its place. Meta never loses the first part, in any setting; all of its losses are the second part. Gemini does the opposite: with automatic detection, it loses the first part in 54% of EN + X recordings and the second in 19%, and the lost part is the English one in 37 of the 100. GPT-Live loses a part in 6–7% of EN + X recordings, almost always the second. Qwen also loses the second part more often than the first (14% against 6% of EN + X recordings with both languages selected, and 101 against 37 of the 300 X + Y); in 14 of its 20 EN + X losses it writes other text in the lost part's place. Whisper SH usually leaves a gap: with both languages selected, it loses the first part in 25% of recordings and the second in 20%, and in 43% nothing is written where the part should be.

Romanised and translated parts leave no gap. Deepgram rarely leaves a gap. Instead, in 34% of the 50 EN + X recordings whose other language has its own writing system (Arabic, Hindi or Mandarin), it writes that part in Latin letters, mostly Arabic and Mandarin parts as strings of similar-sounding Latin syllables. Gemini, with both languages selected, writes the Hindi part in Latin letters in 5 of the 10 recordings of English and Hindi. A real English translation is mainly a Whisper behaviour, as reported for code-switched speech [38]: Whisper SH, with only English selected, translates the other part into English in 18% of these recordings. A check for missing text does not find romanised or translated parts.

One language per session, and finalising once. Zoom took one language per session, so all three settings behave like English only: it loses a language in 62% of EN + X recordings and 85% of X + Y. Its Arabic, Hindi and Mandarin parts are lost in every recording, while most Spanish, German and French parts survive (43 of 130, 7 of 110 and 19 of 110 lost, with both languages selected). ElevenLabs, by default, finalises the whole recording as one block of text. It loses a part in 62% of EN + X recordings (almost always the non-English part) and 93% of X + Y; ElevenLabs NV in 80% and 96%. Set to finalise at each pause of 300 ms, ElevenLabs lost a part in 1% of the screened EN + X recordings and 36% of the screened X + Y (Appendix G). This pause matches the one we put between the two clips, so it favours this setting.

9 Before the Model: A Case Study of Running Whisper

Whisper is a widely used open ASR model. Unlike the hosted services in this paper, it leaves several choices to whoever runs it: where it is hosted, how the audio is cut into segments, and whether a VAD drops audio before the model hears it. On whispered speech, these choices changed our results more than the choice between most systems did. This section describes what we tried and what each choice did in our test.

The nine setups. Table 10 lists them. Setups 1–6 run Whisper Large V3 on two deployments of ours. OpenAI does not say which Whisper version it serves for setups 7–9; at its 2023 launch it described its Whisper API model as Large V2 [35], so differences between setups 1–6 and 7–9 may come from the model version as well as from how it is run. Appendix H gives the exact rules of each setup.

	Setup	How the audio is cut into segments	Audio dropped before the model
1	Gated, VAD on ("gated Whisper")	server's own speech detection	Silero VAD on the client drops silence
2	Gated, VAD off	server's own speech detection	none
3	Self-hosted, Silero cuts ("Whisper SH")	client, at pauses Silero finds	none
4	Self-hosted, Silero drops	client, at pauses Silero finds	Silero drops silence
5	Self-hosted, TEN cuts	client, at pauses TEN VAD [53] finds	none
6	Self-hosted, TEN drops	client, at pauses TEN finds	TEN drops silence
7	RT Whisper	OpenAI's streaming service	none
8	Whisper-1	none: the whole recording is uploaded at the end	none
9	Whisper-1 split	client, at pauses Silero finds	segments with no speech are not sent

Table 10: The nine ways of running Whisper. Setups 1–2 and 3–6 are two different deployments of Whisper Large V3; setups 7–9 are OpenAI's hosted Whisper.

Table 11 compares the nine on the same clips.

	Gated VAD on	Gated VAD off	Self-hosted Silero cuts	Self-hosted Silero drops	Self-hosted TEN cuts	Self-hosted TEN drops	OpenAI RT Whisper	OpenAI Whisper-1	Whisper-1 split
English, quiet	8.7	9.5	8.3	8.0	7.8	7.2	8.2	**5.5**	6.5
English, fan on	7.0	7.5	6.5	5.3	5.9	5.9	**4.6**	5.3	6.3
English, whispered	80.7	61.1	10.1	51.4	10.2	10.7	8.9	**7.7**	23.4
10 other languages†	17.7	17.3	18.4	18.6	18.4	17.6	**14.0**	18.8	18.9
Hindi	29.4	30.3	29.8	32.9	29.8	31.6	**11.8**	36.0	36.0
Spanish (Spain)	10.0	10.0	19.5	14.2	19.5	14.7	**7.9**	8.4	8.4
English, quiet, formatted	16.1	17.5	16.0	16.0	15.3	15.5	11.9	**10.1**	13.4
Wait, median (ms)	395	**360**	361	385	390	395	735	1,850	1,550
Wait, 90th percentile (ms)	624	596	**498**	540	533	538	832	4,105	2,314
No-speech clips with text (of 10)	**0**	**0**	6	**0**	6	2	**0**	10	**0**
Lost a language, EN + X (%)	34	–	45	53	39	54	**11**	89	68
Lost a language, X + Y (%)	60	–	87	96	93	96	**11**	94	91

Table 11: Nine ways of running Whisper for dictation. Error rates in %, lower is better; the lowest in each row is in bold. †Average over 10 language sets, not counting Urdu. Lost a language: share of all two-language recordings (100 EN + X and 300 X + Y) in which one language went missing, with both languages selected. Gated Whisper ran the two-language test with its VAD on only.

Finding 1: Silero VAD dropping audio lost whispered speech. With its VAD on, the gated deployment has a whispered WER of 80.7%, and the VAD never sent 2 of the 16 whispered clips. Turning the VAD off lowers whispered WER by 19.6 points (95% CI 7.0 to 36.9) and changes quiet and fan-on English by less than a point (Appendix H, Table 26). The self-hosted deployment shows the same effect from the other side: sent all of the audio, its whispered WER is 10.1%; with Silero dropping audio, 51.4%, with 44.6% of words missing. Both effects hold across the 107 EARS speakers: turning the gated deployment's VAD off lowers whispered WER by 10.1 points (95% CI 8.0 to 12.2), and Silero dropping audio raises the self-hosted deployment's by 26.8 points (95% CI 21.3 to 32.2), with 62 of the 107 speakers at 20% or more, against 20 (Table 27).

Finding 2: TEN VAD dropped audio with a much smaller loss. When the VAD only cuts the audio, Silero and TEN give nearly the same results (whispered 10.1% and 10.2%). When it also drops audio, they differ sharply: with Silero, whispered WER rises to 51.4%; with TEN, it stays at 10.7%. On EARS, dropping with Silero gives 38.3% and dropping with TEN 14.4%, against 11.5% for sending all of the audio; TEN's rise of 2.9 points is within chance (95% CI −0.4 to 6.7). TEN keeps whispered speech because it calls most of it speech: 79% of whispered 32 ms windows at its default threshold, against 37% for Silero. TEN does not do this by calling everything speech: on fan noise, steady noise, mains hum and television it calls about as much audio speech as Silero, and at most 2.2% of the windows of any no-speech clip. It does call much more background talk speech: with babble, 97% of windows against 80% for Silero, 15 points more than on the same recordings without babble. We did not test what Whisper writes when TEN passes that audio.

Finding 3: lowering Silero's threshold did not fix it. With Silero's drop threshold lowered from 0.5 to 0.2, whispered WER still stayed at 41.1%, four times the 10.1% of sending all of the audio, because even at 0.2 Silero calls only 52% of the whispered audio speech (Appendix H, Figure 2). TEN keeps whispered WER between 10.0% and 10.7% at every threshold tried.

Finding 4: Whisper wrote invented text on silence. Whisper can write text that was never said (hallucination) [23; 31]. On all 10 no-speech clips Whisper-1 wrote text, mostly "you" and "Thank you for watching.", and the self-hosted deployment wrote a stray "." or "so" on 6 of 10. On two short Spanish (Spain) clips the self-hosted deployment replaced the whole speech with a stock phrase such as "Muchas gracias." Dropping audio with a VAD removes text on silence, at the cost above.

Finding 5: the most accurate setups were the slowest. Whisper-1, which transcribes the whole recording after the user stops, has the lowest quiet and whispered English WER of the nine setups (5.5% and 7.7%), but a typical wait of 1,850 ms and text on every no-speech clip. RT Whisper is the most accurate on the other languages (14.0% average, against 17.3–18.9%) and loses a language in 11% of two-language recordings, but accepts only one language and waits about twice as long as the self-hosted deployment (735 ms against 361 ms typical).

Finding 6: cutting at the speaker's pauses kept whispered words. With its VAD off, the gated deployment still has a whispered WER of 61.1%, almost all of it missing words, while the self-hosted deployment, which transcribes exactly the segments the client cuts at pauses, loses 1.8% of whispered words on the same clips and 10.4% on EARS. The two are different deployments, and the gated server runs its own speech detection, so this comparison does not isolate how the audio is cut (Appendix H).

10 Recommendations for Testing ASR for Dictation

Systems with similar error rates made different kinds of errors in our test, at every level of the taxonomy. We recommend that tests of ASR systems for dictation include the following.

- **Recognition: report wrong, missing and extra words separately**, with missing stretches and cut-off endings, because a cut-off ending leaves no visible gap (Sections 6.1 and 6.2). Test whispered speech with several speakers. Report accuracy on names separately, because a gap of 1 point and one of 10 look almost the same in overall WER, and check which words are names (Section 6.3).
- **Output formatting: say which reference is used**, and score against both what the speaker meant and what they said when clean-up settings are compared (Section 7.3).

Check how numbers are written (Section 7.1) and the writing system of the output, especially for languages written in more than one (Section 7.2).

- **Language handling: test two-language speech under each language setting**, and tell missing, romanised and translated parts apart (Section 8). Settings that seem unrelated to language can matter: ElevenLabs lost far fewer languages when it finalised text at each pause.
- **Before the model: test the whole app**, including any VAD and how the audio is cut, on whispered speech and on silence (Section 9).
- **Separate errors all systems share from errors one system makes.** Errors every system makes, such as on Saudi and Emirati Arabic, will not be fixed by switching among the tested systems (Section 6.4).

11 Limitations

- **One speaker.** The in-house English is 106 scripted messages read by one speaker, who varies accent deliberately but has one voice. It is read speech, not spontaneous dictation, so it has fewer and different disfluencies than real dictation, which matters most for the clean-up results.
- **Small sets.** There are 16 whispered messages and 20 clips per language set, so many counts are small and the tables show patterns rather than exact rates. Each message was recorded in only one condition, so the quiet, fan-on and whispered sets differ in their messages as well as their condition.
- **Corpora and sources.** Each dialect set comes from a single corpus, so dialect and recording source cannot be separated. The external corpora are read speech too, and their texts are public (EARS uses one well-known passage, VOiCES uses LibriSpeech [36]), so a system may have seen them in training; we cannot check this, and it could flatter their clean scores and make the whispered EARS test easier for systems that remember the passage.
- **Streaming.** We stream the audio and measure the wait for the final text, but we score only finalised text: we do not measure partial text or how it changes [46], or how well a system decides by itself that the speaker has stopped.
- **Hosted systems change.** Results for hosted services reflect their versions in October 2026, tested on different days, and each clip was played once per system and setting except where stated.

Appendix I lists further limitations: how the references and labels were made, choices made on the test data, systems and settings not fully tested, and timing.

12 Conclusion

We streamed the same audio to eleven ASR systems and grouped every error by the level it shows up at. Systems with similar error rates failed differently at each level. In recognition, three systems left out about a quarter of whispered words that the others heard, whatever their end-of-turn settings, a pattern that largely held across 107 external speakers, and names were written correctly less often than other words, by up to 9.7 points, though only Deepgram's gap is clear after correcting for the number of comparisons. In output formatting, systems differed in how they wrote numbers and in which writing system, and a clean-up setting wrote the message meant rather than every word said. In language handling, selecting only English removed the

other language for some systems and not for others. Before the model, Silero VAD dropping audio raised Whisper's whispered WER from 10.1% to 51.4%, while TEN VAD did not clearly raise it, and the default speech detection of our Qwen deployment cost it most of its whispered words. Some errors, such as those on Saudi and Emirati Arabic, are made by every system. We recommend reporting these error types alongside WER and testing the app's whole audio path.

Ethics

The English dictations were recorded by one member of our team, who volunteered to read scripted messages for this study and gave consent to the use of their recordings for it and to their public release; the consent is kept on record by the authors. The scripts were written for the study, and the people they name are invented, except that three email scripts are signed with the name of one of the authors, who agreed to this; the scripts contain no other personal data. No audio or text from users of any product is included. Because the test streams audio to hosted ASR services, the recordings were processed by those providers under their terms of service. The public clips are used under their licences. EARS (CC BY-NC 4.0) and VOiCES (CC BY 4.0) are used for non-commercial research evaluation only, kept apart from our own recordings and not redistributed. No data from this study, and no transcript returned by a tested system, was used to train any model; the systems' transcripts were used only to score them and to draft the spoken references (Section 4).

Code and data availability. The test client, the scoring and analysis code, and the scripts that build every table in this paper are released at `https://github.com/heylemon-ai/streaming-dictation-asr` under the Apache 2.0 licence. The in-house English recordings (quiet, fan on and whispered, and the no-speech clips), both references for each, and every tested system's transcripts are released at `https://huggingface.co/datasets/hey-lemon/streaming-dictation-asr` under CC BY 4.0. The other language sets, the two-language recordings, EARS and VOiCES are not redistributed. The dataset lists the public clips used and the recipe for the two-language recordings; the repository has scripts that fetch EARS and VOiCES from their sources and rebuild the two-language recordings once the public clips are in place.

Competing interests. The authors work at a company that makes dictation software and is developing its own ASR model, and that has commercial relationships with some of the tested providers. The authors and their company have no partnership with any of the tested providers. This work received no external funding: it was carried out and paid for by the authors' company, and no tested provider funded it.

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A Further Related Work

Latency and the written form. In streaming ASR, the delay a user feels depends on how quickly text is emitted and on endpointing, the decision that the speaker has stopped [47]. The written form is usually produced after recognition: inverse text normalisation (ITN) turns spoken forms such as "seven" into "7" [40], and disfluency removal turns a verbatim transcript into fluent text [20]. We score the output of these steps as the user receives it. Streaming systems also show partial text that may change before it is final; how often it changes can be measured [46], but we score only the final text.

Running Whisper. Whisper processes audio in 30-second windows, so live use needs extra steps. Whisper-Streaming re-runs the model on a growing buffer [30], and WhisperX cuts long audio with a VAD before transcription [6]. Whisper can also write text that was never said, often around silence [23] and in noisy audio [31]. Section 9 compares such choices for dictation.

B Systems and Settings

We use each system's default settings, with the changes below. For Deepgram we turn on `punctuate` and `smart_format`, which add punctuation and change how numbers, dates and similar items are written. GPT-Live has a `delay` setting that trades how early partial text appears against accuracy, and we found no documented default. We use `low`, first chosen on 50 of the English test clips.

GPT-Live's delay setting. To check that choice, we ran `low` and `minimal` on the same day on every test set (Table 14). The typical wait for the final text is the same (744 and 752 ms; slow case 899 and 882 ms), and neither writes text on any no-speech clip. `minimal` has the higher WER on every English set, ours and the external ones, and on 9 of the 11 other language sets. Treating the 25 comparisons of the table as one group, the difference is clear (Holm-corrected) on quiet and whispered English, on both EARS sets and on three of the four VOiCES conditions, and of the other languages only on Hindi. It loses a language as often on EN + X recordings, and more often on X + Y with automatic detection (8.3% against 5.0%, also clear). The `low` runs of this check match those in the main tables to within 0.6 points on EARS and VOiCES; on our sets, Spanish (Puerto Rico) moved most (5.8% in the main run, 3.1% on 5 November), then Hindi (11.8% and 14.0%).

Other settings. RT Whisper accepted only one language in our tests, so with two languages selected it is told neither. Whisper SH decodes with beam search (width 5); with two languages selected, it is given both as the allowed languages. It does no segmentation of its own, so the

client cuts the audio into segments (Section 9). For ElevenLabs, the default settings mean the text is verbatim, including fillers such as "uh", and the text is finalised once, when the audio ends; Section 8 also tests finalising at each pause. For Zoom, we use the settings of its sample app, with text clean-up turned off. Each system is told the language a user would choose. If a system has no option for that language, we send none and the system detects the language itself. Zoom has no locale for Egypt, Latin America or Puerto Rico, so for those sets we use the closest one it has (Saudi Arabic; Spanish, Spain). It has no option for Urdu at all.

Regional language codes. Systems that take a region with the language get one code per language in the main runs: Gemini `ar-EG` for every Arabic set and `es-US` for every Spanish set, and Zoom `ar-SA` and `es-ES`; the others take the bare language. On 5 November we ran each set again with the regional codes the providers document: Deepgram with `ar-EG`, `ar-SA`, `ar-AE` and `es-419`; Zoom with `ar-AE`, which its service accepted; and Gemini with `es-419` (its transcription documentation lists no Saudi or Emirati Arabic code). The transcripts were the same as in the main run on 78 of Deepgram's 80 clips, all 20 of Zoom's and 33 of Gemini's 40, and no failed clip changed. The largest change in WER was Deepgram on Spanish (Latin America), from 8.7% to 7.8%. So the dialect results do not come from the codes we sent.

End-of-turn and speech-detection checks. On 5 November we ran the 16 whispered clips again with four changed setups, each against the system's main-run results: AAI 3.6 and AAI 3.5 sending `ForceEndpoint` before `Terminate`, so the turn in progress is finalised; the same with end-of-turn detection effectively off (`end_of_turn_confidence_threshold` 1, `max_turn_silence` 10 s); Zoom with `silence_duration_ms` 5,000 instead of 600; and Qwen with our deployment's speech detection relaxed (`streaming_vad_config`: threshold 0.3, 2,000 ms of silence, 300 ms of padding), alongside a rerun with its default settings, which matched the main run (32.2% of words missing in both). Section 6.2 gives the results.

Each clip is sent in small chunks at the speed of speech, the way a microphone delivers it. Most systems receive 32 ms chunks at 16 kHz. Meta, GPT-Live and RT Whisper receive 32 ms chunks at 24 kHz, the rate their APIs expect, resampled from the 16 kHz audio. The AAI models and Zoom receive 100 ms chunks: AssemblyAI's API rejected chunks shorter than 50 ms in our tests, and Zoom's sample app uses 100 ms. When the clip ends, the client tells the system the audio is over and waits up to 30 seconds for the final text. Every clip uses a new session, and all systems were called from the same client machine.

To check that the systems did not change during the test, we replayed 42 clips to two hosted services (Gemini and Meta) and to the gated Whisper deployment (described in Section 9) one week after their first run. Whisper and Gemini returned the same text, and Meta's WER barely moved (16.5% against 16.7%, on the 41 clips it answered; one session was refused by a rate limit). The other systems were not rechecked.

Table 12 lists the exact model each system was called with, how it was called and when. All systems were called over their own streaming interfaces (WebSocket), with our own client code rather than a vendor SDK.

System	Model	Interface	Audio sent	Tested (2026)
Whisper SH	Whisper Large V3, our own deployment	WebSocket, client cuts segments	16 kHz, 32 ms	20–21 Oct; 29 Oct (EARS, setups that drop audio or use TEN)
Qwen	Qwen3-ASR 1.7B, our own deployment (one H100 MIG slice)	WebSocket	16 kHz, 32 ms	1–13 Oct; 5 Nov (two languages)
Meta	`muse-voice-transcribe-1.0`	WebSocket	24 kHz, 32 ms	1–12 Oct
Gemini, Gemini Smart	`gemini-3.5-transcribe-live`	Live API	16 kHz, 32 ms	1–15 Oct, 21 Oct
Deepgram	`nova-3`	WebSocket	16 kHz, 32 ms	12 Oct
AAI 3.6, AAI 3.5	`universal-3-6-pro`, `universal-3-5-pro`	WebSocket	16 kHz, 100 ms	12 Oct
Zoom	Zoom Scribe (Live API)	WebSocket	16 kHz, 100 ms	14 Oct
ElevenLabs, ElevenLabs NV	`scribe_v2_realtime`	WebSocket	16 kHz, 32 ms	14–15 Oct; 29 Oct (finalise at each pause)
GPT-Live	`gpt-live-transcribe`, `delay: low`	Realtime API	24 kHz, 32 ms	19 Oct
RT Whisper	`gpt-realtime-whisper`	Realtime API	24 kHz, 32 ms	20 Oct
Gated Whisper	Whisper Large V3, our deployment with a client VAD gate	WebSocket, server cuts segments	16 kHz, 32 ms	1–21 Oct; 29 Oct (EARS, VAD off)

Table 12: Models, interfaces and test dates of the main runs. Every row was also run on EARS on 22–23 October and on VOiCES on 23–24 October. Our own deployments run on Baseten; their deployment identifiers are not given. Tests ran from one client machine.

Table 13 gives the settings each system was called with: how the language was set, what ended the audio, and when the system finalised text. Anything not listed was left at the provider's default.

System	Language setting	End of audio	When text is finalised	Other settings
Whisper SH	`audio_language`; two languages as `language_options`	the client cuts the last segment	each segment the client cuts (Section 9)	beam width 5; partial text off
Gated Whisper	`audio_language`	`end_audio`	server's own speech detection (threshold 0.7, 480 ms silence)	beam width 5; partial text off; client VAD in front
Qwen	`audio_language`	`input_audio_buffer.commit`	server	partial text off
Meta	`languageBias`	`endStream`	once, at the end	push-to-talk mode; cumulative partial text
Gemini, Gemini Smart	`languageCodes`	`activityEnd`	once, at the end (automatic activity detection off)	`mode`: `VERBATIM` or `SMART`
Deepgram	`language`; `multi` for two languages or none	`CloseStream`	server's endpointing, at its default	`punctuate`, `smart_format`
AAI 3.6, AAI 3.5	`language_codes`	`Terminate`	server's turn detection, at its default	`format_turns`
Zoom	one locale per session	`session.close`	server's turn detection: 600 ms of silence, at most 30 s	partial text off; text clean-up off
ElevenLabs, ElevenLabs NV	`language_code`; second language as `secondary_languages`	a final empty chunk with `commit`	once, at the end (`commit_strategy`: `manual`), though the server finalised the first 36 s of longer messages itself (Table 22); the *pause* runs: at each 300 ms pause (`vad`)	NV: `no_verbatim`
GPT-Live	`languages`	`input_audio_buffer.commit`	once, at the end (server turn detection off)	`delay`: `low`
RT Whisper	`language` (one only)	`input_audio_buffer.commit`	once, at the end (server turn detection off)	–

Table 13: Settings sent to each system. Parameter names are the providers' own. Partial text, where a system sends it, was not used: only finalised text is scored.

	`low`	`minimal`	Difference [95% CI]
Our sets, WER			
Quiet	7.4	9.3	+1.8 [1.3, 2.3]
Fan on	3.8	5.6	+1.8 [0.3, 3.4]
Whispered	8.8	13.6	+4.9 [2.4, 7.1]
Arabic MSA (Egypt)	7.1	8.2	+1.0 [−1.5, 3.4]
Saudi Arabic	27.3	32.8	+5.5 [1.4, 10.6]
Emirati Arabic	53.5	57.1	+3.5 [−1.9, 8.8]
German	6.5	7.7	+1.2 [−2.8, 4.9]
Spanish (Latin America)	6.5	8.7	+2.2 [−0.9, 5.9]
Spanish (Spain)	8.4	12.1	+3.7 [0.5, 7.6]
Spanish (Puerto Rico)	3.1	6.8	+3.7 [0.0, 9.2]
French	9.4	14.5	+5.1 [1.5, 8.6]
Hindi	14.0	20.6	+6.6 [3.1, 10.2]
Urdu	21.5	19.7	−1.8 [−14.3, 9.0]
Mandarin	7.6	7.1	−0.5 [−2.8, 1.6]
EARS and VOiCES, WER			
EARS, normal	1.3	2.0	+0.7 [0.5, 0.9]
EARS, whispered	2.1	4.0	+1.9 [1.5, 2.3]
VOiCES, clean	3.5	4.7	+1.3 [0.8, 1.8]
VOiCES, far	8.0	12.3	+4.3 [3.2, 5.4]
VOiCES, far + babble	84.5	85.8	+1.4 [0.2, 2.8]
VOiCES, far + TV	29.4	35.8	+6.4 [4.7, 8.4]
Two languages, share of recordings with a language lost			
EN + X, both selected	6.0	6.0	+0.0 [−3.0, 3.0]
EN + X, English only	7.0	7.0	+0.0 [−3.0, 3.0]
EN + X, auto-detect	7.0	7.0	+0.0 [−3.0, 3.0]
X + Y, both selected	4.3	5.0	+0.7 [−0.7, 2.0]
X + Y, auto-detect	5.0	8.3	+3.3 [1.3, 5.3]

Table 14: GPT-Live with its `delay` set to `low` (used in every other table) and to `minimal`, run on the same day: 5 November for our sets, EARS and VOiCES, and 19 October for the two-language recordings. WER (%), scored as in Table 3 for our sets and with Whisper's English normaliser for EARS and VOiCES (Appendix F); two-language rows are the share (%) of all recordings in which a language went missing. Difference: `minimal` minus `low`, from unrounded values, with a paired bootstrap interval over clips (our sets), speakers (EARS and VOiCES) or recordings (two languages).

C Data and Scoring Details

The spoken reference. To build the spoken reference, we aligned the transcripts of the eleven systems' default (verbatim) settings with the script and proposed an edit wherever several systems heard the same departure (an added word when at least three did; a missing or changed word when at least five of eleven did, because on whispered speech four systems drop the same quiet words). Proposals with stronger agreement (at least four, or at least six) were shown already ticked. A member of the team then listened to every message, accepted or rejected each proposal and corrected the text by hand where needed (six messages). Complete words were kept as spoken, including words the reader then corrected; cut-off fragments and fillers ("um", "uh") were left out, as they are not in the scripts either.

Correcting for many comparisons. A difference is called clear only if its paired 95% interval excludes zero and its two-sided paired-bootstrap p-value stays below 0.05 after a Holm correction within its group. The groups are: the lowest against the next-lowest system in each English condition; the four whispered missing-word shares against 3%; the 13 name gaps; Deepgram's name gap against each other system's; script against spoken reference on quiet English, and the

two clean-up settings against each other on the spoken reference; the two missing- and extra-word differences between Whisper SH and AAI 3.6; ElevenLabs NV against ElevenLabs on the Saudi and Emirati Arabic sets; gated Whisper with its VAD on and off; the Whisper setups on in-house whispered speech; English only against both languages selected, on all recordings; ElevenLabs finalising at each pause against once; the EARS changes from normal to whispered; the EARS Whisper setups; the VOiCES changes from the quiet far recording, with babble and with TV; and GPT-Live's two delay settings (Table 14). The scripts `paper/analysis/stats_review.ts` (4,000 resamples) and `gptdelay.py` (2,000) compute them, and for the two-language tests the bootstrap resamples source clips (Section 8). A difference shown only by its interval, outside these groups (for example the language averages and the name gaps on quiet English alone), is not called clear.

Normalisation. Before comparing a transcript with the correct text, we ignore capital letters and punctuation. For Arabic and Urdu we remove short-vowel marks and the elongation character; for Arabic we also treat *alif* with and without *hamza* as equal, *ta marbuta* as *ha*, and *alif maqsura* as *ya*. We do not merge the Arabic and Urdu forms of *yeh*, *kaf* and *heh*; no Urdu reference or transcript uses the Arabic forms, so this changes no score. Traditional and simplified Chinese characters are treated as equal.

The language sets. Most are sentences read aloud (FLEURS, and the Puerto Rican Spanish set); the Arabic MSA (Egypt) set is Modern Standard Arabic read by Egyptian speakers, while the Saudi and Emirati sets are dialect speech, and the Spanish (Spain) set is European Parliament speech.

Mandarin. Mandarin is not written with spaces, so we count each Chinese character, and each run of Latin letters or digits, as one unit: the mixed error rate (MER) used in Mandarin–English code-switching work [29], not the match error rate of Morris et al. [34]. We also compute the character error rate (CER), the same measure counted over characters; on our single-language Mandarin set MER equals CER. Chinese numerals in place of digits count as errors, as numbers do in every language.

Labelling names. Each different word was then labelled as a name or not, following a written rule: part of the name of a person, organisation, product, service, programming language, place or titled work counts; days, months, time zones, titles such as "Dr.", job titles, acronyms such as "API" or "PM", document headings and ordinary words written with a capital do not. Only 156 of the 327 capitalised words are names, so the capital-letter rule alone is a poor way to find them. Adding the 22 occurrences of a listed name that the capital-letter rule misses (at the start of a sentence, or after "Dr." or "Mr.") and the 5 lower-case uses of a listed product name gives 183 name words (156 + 22 + 5).

D Accent Labels

We labelled each in-house recording with CommonAccent [59], an English accent classifier with 16 classes trained on Common Voice, over 4-second windows taken every 2 seconds, keeping windows whose top class has a probability of at least 0.7. On the 90 spoken messages (quiet and fan on), 62% of these windows are labelled US English and 29% South Asian English (the classifier's Indian English class); England, New Zealand and Ireland together make up 7%, and every other class under 1%. South Asian English appears in 64 of the 90 messages but is the label for the whole message in only 18. Of consecutive windows, 28% change label, against 4% on the 428 normal-voice EARS clips, whose 107 speakers each keep one accent; we take the EARS rate as the classifier's own noise. The reader confirmed that the US, South Asian, British and

Irish stretches were deliberate. On whispered messages the classifier labels 98% of windows US English and changes label no more often than on EARS, so it finds no accent cue in whispered speech. Messages labelled mostly South Asian English (25 of the 90) have about the same WER as the rest for every system in Table 3: no difference is clearly outside chance in a bootstrap over messages. But 16 of the 25 are the fan-on messages, so accent and condition cannot be separated.

E Additional Results

Results by language. Table 15 gives WER for each language set. Counting Urdu changes the order of the averages. Without it, Gemini has the lowest average (11.8%); in a paired bootstrap over clips, its 95% interval against every system except Gemini Smart (12.4%) and AAI 3.6 (13.3%) excludes zero, but these 12 comparisons are not corrected for their number, so we do not call them clear. With it, AAI 3.6 (13.5%), RT Whisper (13.9%), Gemini Smart (14.2%), GPT-Live (14.5%) and Gemini (15.4%) are within chance of each other.

System	Arabic MSA (Egypt)	Arabic Saudi	Arabic Emirati	German	Spanish Lat. Am.	Spanish Spain	Spanish P. Rico	French	Hindi	Mandarin (MER)	Urdu	Avg. no Urdu	Avg. with Urdu
Whisper SH	9.2	37.1	60.0	6.9	5.7	19.5	5.8	6.5	29.8	4.1	19.7	18.4	18.6
Qwen	7.7	39.1	61.2	4.4	7.4	8.9	8.4	7.2	18.4	3.5	94.7	16.6	23.7
Meta	6.6	27.7	49.4	6.5	5.2	10.5	5.2	8.7	14.9	6.5	87.3*	14.1	20.8
Gemini	2.6	29.3	48.8	5.2	6.5	8.4	3.7	7.2	3.1	3.3	51.8*	11.8	15.4
Gemini Smart	3.6	35.2	49.4	4.8	5.7	7.4	4.7	6.9	2.6	3.8	32.4*	12.4	14.2
Deepgram	11.2	38.3	57.1	8.1	8.7	9.5	4.2	10.5	20.6	3.3	14.4	17.1	16.9
AAI 3.6	9.7	33.6	53.5	4.8	2.2	9.5	5.2	2.9	9.2	2.7	15.1	13.3	13.5
AAI 3.5	14.3	34.8	57.1	6.0	5.2	10.0	5.2	4.0	25.9	3.5	91.5*	16.6	23.4
Zoom	5.1	29.7	51.8	4.8	6.1	10.0	4.2	5.1	34.6	3.3	102.5*	15.5	23.4
ElevenLabs	10.2	34.8	61.8	4.0	2.6	8.9	8.9	9.1	11.8	5.2	22.2	15.7	16.3
ElevenLabs NV	9.2	66.0	79.4	3.6	1.7	7.9	8.4	9.4	10.5	4.4	19.4	20.1	20.0
GPT-Live	6.6	27.7	51.8	4.4	6.5	8.9	5.8	9.4	11.8	6.5	19.7	14.0	14.5
RT Whisper	7.1	28.9	55.9	3.2	3.9	7.9	7.3	7.2	11.8	6.8	12.3	14.0	13.9

Table 15: WER (%) for each language set (20 clips each), lower is better. Nothing is in bold: in a paired comparison, the lowest WER in each set is never clearly below the next one. Arabic MSA (Egypt): Modern Standard Arabic read by Egyptian speakers. Mandarin uses the mixed error rate. Avg.: unweighted mean of the per-set WER over the 10 sets other than Urdu, and over all 11 with Urdu; the second counts each system's Urdu output as returned, including systems with no Urdu option, because that is what a user choosing Urdu receives. *The system has no option for this language and detects it itself; not ranked.

How whispered dictations fail. Table 16 shows how many whispered clips failed and where the missing stretches fall.

		Clips with WER		Missing stretch	Cut-off
	WER	≥50%	≥90%	at end	ending
Whisper SH	10.1	0	0	0	0
Qwen	46.6	38	25	6	12
Meta	11.3	0	0	0	0
Gemini	5.0	0	0	0	0
Gemini Smart	4.2	0	0	0	0
Deepgram	19.8	0	0	6	0
AAI 3.6	30.5	31	19	12	25
AAI 3.5	30.9	31	19	12	25
Zoom	34.7	31	12	6	0
ElevenLabs	9.1	0	0	0	0
ElevenLabs NV	7.1	0	0	6	0
GPT-Live	8.8	0	0	0	0
RT Whisper	8.9	0	0	0	0

Table 16: How whispered dictations fail (16 clips); all values are percentages. Figure 1 splits the WER into wrong, missing and extra words. Clips with WER: share of clips with at least that WER. Missing stretch at end: share of clips with 5 or more words in a row missing in the last third. Cut-off ending: share of clips whose last 3 or more words are missing.

Number habits and normalised scores. These switches are part of the WER differences in Table 3, and a benchmark that converts numbers to a common form does not measure them. To see how much they matter, Table 17 re-scores English after applying Whisper's English text normaliser [41] to both the reference and the transcript. The normaliser writes numbers as digits, standardises contractions and spellings, and removes fillers such as "uh". The normalised scores also split text into words slightly differently, so we compare them with the raw WER under the same word splitting. On quiet English every WER falls, by 1.0–3.8 points. The systems that write digits as words or keep fillers gain the most: Qwen (3.8 points), ElevenLabs (3.5), Deepgram (2.5), Meta (2.2), and GPT-Live and AAI 3.5 (2.1 and 1.6). The order changes little (Spearman rank correlation 0.98), and Gemini Smart stays lowest on quiet English. An app that converts numbers after recognition would remove part of this difference; we did not test this.

System	Quiet	Fan on	Whispered
Whisper SH	7.3	5.6	7.7
Qwen	10.2	4.6	43.6
Meta	7.4	4.3	7.6
Gemini	6.3	1.8	3.3
Gemini Smart	3.4	3.3	3.5
Deepgram	11.2	9.7	16.4
AAI 3.6	6.8	3.5	29.4
AAI 3.5	7.4	3.8	30.1
Zoom	6.0	2.6	33.8
ElevenLabs	9.8	4.4	7.1
ElevenLabs NV	4.3	3.3	5.6
GPT-Live	5.7	2.1	6.7
RT Whisper	7.0	3.6	6.7

Table 17: English WER (%) after Whisper's English text normaliser, which writes numbers as digits, standardises contractions and spellings and removes fillers, applied to reference and transcript alike. Tokenisation differs slightly from Table 3.

Gemini Smart against Gemini. Smart lowers Gemini's quiet-English WER from 6.9% to 3.9%. Almost all of the difference is in extra words, which fall from 3.7% to 0.7% of words.

With the fan on (3.2% against 2.5%) and whispered speech (4.2% against 5.0%) the two settings are close. When capital letters and punctuation are also scored, Smart is worse with the fan on (10.0% against 6.4%) and whispered speech (10.9% against 8.6%). In the other languages, the largest changes are Urdu, where WER falls from 51.8% to 32.4% (Section 7.2), and Saudi Arabic, where it rises from 29.3% to 35.2%.

ElevenLabs NV against ElevenLabs. Without verbatim output, ElevenLabs' quiet-English WER falls from 12.8% to 5.1%, and its extra words from 8.1% to 1.4% of words. With the fan on (4.5% against 5.8%) and on whispered speech (7.1% against 9.1%) it also improves. The Saudi and Emirati Arabic sets get clearly worse: Saudi Arabic rises from 34.8% to 66.0% (paired difference 31.3 points, 95% CI 21.5 to 40.2) and Emirati Arabic from 61.8% to 79.4% (17.6 points, 95% CI 9.1 to 25.9), while Arabic MSA (Egypt) and the other languages change by less than 3 points (Table 18 gives its failed clips).

Ranks against the two references. Apart from the top of the quiet column, the order of the systems changes little: over the thirteen rows of Table 8, the Spearman rank correlation between the two columns is 0.93 on quiet English, 0.78 with the fan on and 0.98 on whispered speech.

Failed clips by language. Table 18 counts each system's failed clips in each set.

System	Saudi	Emirati	Hindi	Urdu	Other	Total
Whisper SH	25	60	20	0	2	11
Qwen	30	75	10	95	1	20
Meta	20	50	15	85	1	16
Gemini	20	55	0	45	1	12
Gemini Smart	20	55	0	25	1	10
Deepgram	30	70	10	0	1	10
AAI 3.6	30	60	5	0	1	10
AAI 3.5	35	70	25	90	1	21
Zoom	15	55	20	100	1	18
ElevenLabs	25	70	10	5	1	10
ElevenLabs NV	75	80	5	0	1	15
GPT-Live	15	55	5	5	1	8
RT Whisper	20	55	10	0	2	9

Table 18: Share of non-English clips (%) with a WER of 50% or more, by set (20 clips per set). Other: the other seven sets together. Total: all 220 non-English clips.

Clips hard for one system only. Across all 326 speech clips, 23 are above 30% for exactly one system: 9 for Qwen, 6 for Deepgram, 3 each for Zoom and ElevenLabs, and 2 for Whisper SH (both Spanish (Spain), where it replaced the speech with a stock phrase; Section 9).

Character error rates on the Arabic dialects. Character error rates are far below the word error rates (11–27% on Saudi and 15–27% on Emirati Arabic with default settings; ElevenLabs NV 41% and 51%), which is consistent with spelling variation but does not show it: Arabic words carry attached prefixes, so CER is far below WER on Arabic MSA (Egypt) too (0.5–4.3% against 2.6–14.3%).

Hindi. AAI 3.5 fails on 25% of Hindi clips, and Whisper SH and Zoom on 20% each. Gemini fails on none, and has the lowest Hindi WER in both settings (3.1%, and 2.6% with Smart). Zoom has the highest Hindi WER of the eleven systems (34.6%). Its text is in the right writing

system, but its character error rate (16.3%) is less than half its WER, so many of its wrong words are near-misses at the character level rather than different words (CER counts Unicode characters, so a vowel sign counts separately). The same holds for Whisper SH (13.6% against 29.8%), and almost for AAI 3.5 (13.0% against 25.9%), but not for Meta or Deepgram, whose character and word error rates are about equal. Merging common Devanagari spelling variants (the nukta, and chandrabindu with anusvara) lowers Hindi WER by at most 2.2 points for any system (Zoom: 34.6% to 33.3%), so these variants do not explain the gap.

What the wrong words are. Table 19 sorts the wrong English words into four groups: a number written the other way ("seven" for "7"), a name, a near-miss (a word of 4 or more letters that differs by 1 or 2 letters, such as a plural or another form of a verb: "request" for "requests"), and any other word.

	Wrong words	Share of wrong words			
System	(% of words)	Number	Name	Near-miss	Other word
Whisper SH	3.6	14	5	23	58
Qwen	5.7	31	4	14	52
Meta	4.0	30	4	19	46
Gemini	2.1	24	5	22	48
Gemini Smart	1.8	30	4	19	47
Deepgram	6.6	15	4	18	63
AAI 3.6	2.8	27	3	23	47
AAI 3.5	2.9	26	4	19	52
Zoom	2.8	18	4	21	57
ElevenLabs	3.0	18	4	22	56
ElevenLabs NV	2.7	22	3	22	52
GPT-Live	3.3	35	5	14	46
RT Whisper	3.3	25	5	20	50

Table 19: What the wrong words are, in all English dictation (quiet, fan on and whispered together). Wrong words: share of all words. The other columns: share (%) of the system's wrong words in each group. Number: a number written in digits where the script has words, or the other way round. Near-miss: a word of 4 or more letters that differs from the correct word by 1 or 2 letters.

Number habits. Table 20 shows how often each system keeps each number form and how often it switches it.

	Digits in script		Words in script		Share of
System	kept	as words	kept	as digits	errors
Whisper SH	79	15	81	7	6.1
Qwen	13	68	93	0	10.4
Meta	48	47	93	1	12.9
Gemini	79	19	89	2	8.4
Gemini Smart	86	12	74	12	14.0
Deepgram	47	37	87	4	7.2
AAI 3.6	95	0	47	42	7.2
AAI 3.5	95	0	48	41	6.7
Zoom	79	13	79	10	5.0
ElevenLabs	75	21	88	2	4.8
ElevenLabs NV	75	22	89	1	11.2
GPT-Live	50	45	95	0	16.1
RT Whisper	65	31	92	1	10.2

Table 20: How each system writes numbers in English dictation (%). Share of errors: number-format switches as a share of all the system's English errors. This is a minimum, because a number made of several words counts once. Kept and switched do not add up to 100%: the remainder were misheard or missing.

E.1 Speed

Table 21 gives two delays, measured on the test client. The *wait* is the time from the client's end-of-audio signal to the final text (the final hypothesis), the delay after the speaker stops. Because the client signals the end, the wait does not include endpointing, the time a system would take to decide by itself that the speaker has stopped. The wait ends at the signal each service gives that it has nothing more to send, which differs between services: the final text itself for Meta, Gemini, ElevenLabs, GPT-Live and RT Whisper; the end-of-session message for AAI and Zoom; the closing of the connection for Deepgram; and an end-of-audio message from our own deployments (Whisper SH, gated Whisper and Qwen). For the services that end with a session message, the wait can include a short time after the last text arrived. Runs used up to six sessions at a time per system. The *connect* time is how long it takes to open a session, which an app could hide by connecting early. The table gives the typical value (the median, p50) and a slow case (p90: 90% of sessions were faster than this). From our client location, Meta, Deepgram, Whisper SH, Qwen and both ElevenLabs settings have typical waits under 400 ms; GPT-Live and RT Whisper wait about 750 ms, and the AAI models have the slowest p90, over 1,200 ms. Gemini (584 ms) and Gemini Smart (434 ms) were run on different days; side by side on the same 50 English clips, twice, their typical waits differed by only 10 ms. Only Gemini's wait grows with the length of the clip: its typical wait is 591 ms on English clips under 20 seconds and 1,084 ms on clips of a minute or more (442 ms and 951 ms for Smart). For every other system it changes by less than 100 ms across clip lengths. A repeat run on 29 October also recorded when text was first finalised during speech and the network round trip to each provider (Appendix E, Table 22); there, waits moved by up to 320 ms, so we treat them as indicative.

	Wait (ms)		Connect (ms)	
System	p50	p90	p50	p90
Whisper SH	361	498	837	940
Qwen	388	542	1592	4428
Meta	313	371	1368	1756
Gemini	584	905	674	814
Gemini Smart	434	703	718	1201
Deepgram	357	431	911	1053
AAI 3.6	722	1210	1227	1399
AAI 3.5	727	1218	1228	1370
Zoom	407	616	1406	1655
ElevenLabs	398	493	685	765
ElevenLabs NV	395	459	660	727
GPT-Live	760	858	1207	1374
RT Whisper	735	832	1185	1312

Table 21: Delays over all single-language sessions. Wait: from the end of the audio to the final text. Connect: from opening the connection to the session being ready. p50: median; p90: 90% of sessions were faster.

Text during speech, and the network. The wait in Table 21 is what a user feels after they stop. A second question is whether any text is final (a final hypothesis, as opposed to partial text) while they are still speaking. On 29 October we played the 106 English messages to every system again, recording when the first finalised text arrived and, during the same runs, the round trip from the test client to each provider (Table 22). Whisper SH, Qwen, Deepgram, Zoom and the AAI models finalise text during speech on 83–99% of messages, typically 5–9 seconds in. Meta, Gemini, GPT-Live and RT Whisper finalise only at the end, as set up (Appendix B). ElevenLabs, though set to finalise once, finalised the first 36 seconds early on every message longer than about 35 seconds, and on no shorter message. Partial text was not measured. The round trip explains part of the wait: subtracting it moves Deepgram from seventh to third, but the same six rows stay slowest (both AAI models, Gemini in both settings, GPT-Live and RT Whisper), at 594–716 ms beyond the round trip. Waits on 29 October differed from the main run by up to 165 ms for most systems and 320 ms for Gemini Smart.

System	Final text during speech (%)	First final text (s)	Wait p50 (ms)	Round trip (ms)	Wait minus round trip (ms)
Whisper SH	95	8.5	400	276	124
Qwen	96	7.0	360	276	84
Meta	0	–	433	56	377
Gemini	0	–	739	58	681
Gemini Smart	0	–	754	58	696
Deepgram	99	5.3	522	309	213
AAI 3.6	83	8.7	841	200	641
AAI 3.5	85	8.6	794	200	594
Zoom	84	8.8	429	28	401
ElevenLabs	53	35.9	392	145	246
ElevenLabs NV	52	35.9	376	145	230
GPT-Live	0	–	728	28	700
RT Whisper	0	–	744	28	716

Table 22: Timing runs of 29 October on the 106 English messages. Final text during speech: share of messages with finalised text before the audio ended. First final text: median time from the start of the audio to the first finalised text, over those messages. Round trip: median TCP handshake time to the provider during the runs, about 300 samples each; for services behind a content-delivery network (OpenAI, Zoom, and in part Gemini and Meta) it reaches only the nearest edge server, so it understates the round trip to the model. Sessions during a network outage on the client side, of about 25 minutes, were discarded and replayed.

Gemini under repeated and parallel use. Because Gemini's wait changed from one day to the next, we ran two more tests on 21 October. In the first, three English clips (10, 45 and 91 seconds long) were played to Gemini and Gemini Smart at the same moment, 20 times each. Across the 60 pairs, Smart's typical wait was 17 ms shorter (it was faster in 42 of 60 pairs), and each setting returned the same text in every round that day. The wait grew with the length of the clip: about 450–500 ms for the 10-second clip and 1,040–1,090 ms for the 91-second one. In the second test, we kept up to 50 sessions open at once. No session failed. The typical wait stayed between 510 and 640 ms, while the slow case grew at 25 and 50 open sessions (to 910 ms for Gemini at 25, and 1,080 ms for Gemini Smart at 50). We repeated both tests on 22 October. Up to 25 open sessions, typical waits were again 575–630 ms; but at 50, Gemini's typical wait rose to 1,369 ms (slow case 3,587 ms) and Gemini Smart's to 697 ms (1,255 ms). In the repeated pairs, Smart was faster in 33 of 60, against 42 of 60 the day before, and each setting returned the same text in 90–100% of rounds rather than every round. Waits under load therefore varied from day to day.

F EARS and VOiCES: Details

The EARS speakers. The 107 EARS speakers are 60 women, 43 men and 4 other or not stated, aged 18 to 75, mostly native speakers of American English. On whispered speech Whisper SH loses 10.4% of words as missing words, against 7.3% for each AAI model, where on our speaker it lost far fewer; the intervals of the three changes overlap, so this difference is within chance.

The VOiCES clips. We take 100 sentences, one per speaker (50 women, 50 men), each in four versions: the original clean recording, and the room recording from one studio microphone at the far position in room 4 with no background sound, with babble (several people talking) and with a television playing. That is 400 clips, 26 minutes per version.

Scoring. The VOiCES references are LibriSpeech text: lower case, no punctuation and numbers as words. So for both corpora we apply Whisper's English text normaliser to the reference

and the transcript, as in Table 17, and report only WER. Two kinds of clip have no text and count as empty transcripts: clips that gated Whisper's VAD never sent, and sessions with no transcript within 30 seconds. Only Gemini had the second kind, all on noisy clips. In the one session we traced, on a babble clip, the connection stayed open and Gemini returned "Add, and," for a 15.8-second sentence, 17 seconds after the audio ended; we did not trace the others, so some may have other causes. Intervals resample speakers.

Distance and noise. On the clean recordings every system is between 2.1% and 4.6%. A far microphone in a quiet room adds little for most: under 3 points for Whisper SH, Gemini (in both settings), the AAI models and ElevenLabs (in both settings), 4–6 points for Meta, Deepgram, GPT-Live, RT Whisper and Zoom, and 8–10 for Qwen and gated Whisper. With babble, WER rises for every system to between 38.7% (Zoom) and 94.9% (Deepgram), 31–85 points above the quiet far recording. For eleven of the fourteen rows (the thirteen rows and gated Whisper) the errors are mostly missing words, which may come from the service's own speech detection as well as from recognition; Qwen and ElevenLabs (in both settings) write more wrong words than they miss. Extra words are at most 7.3%, so no system writes down much of the background talk. Gemini returned no text within 30 seconds on 67 of the 100 babble clips and 7 of the television clips, and Gemini Smart on 12 babble clips; these count as empty transcripts, so most of Gemini's babble WER is these sessions. With a television, WER rises less and differs more between systems: Zoom rises by 2.6 points (95% CI −0.4 to 5.8, within chance) and the AAI models by 7.4–8.3, while Deepgram rises by 40 and RT Whisper by 32. Zoom shows that the quiet far-microphone results do not predict the noisy ones: it is worse than average with a far microphone in a quiet room, but lowest with babble and with television. Left out of the WER, Gemini's 67 babble sessions with no text leave a WER of 40.3% on the other 33 clips, and Gemini Smart's 12 leave 67.1%; these are only the easier clips, so they are lower bounds.

System	Clean	Far mic	Far + babble	Far + TV	Far + babble: wrong / missing / extra	Babble: no text
Whisper SH	2.1	4.2	62.5	17.7	17.7 / 41.6 / 3.3	0
Gated Whisper	2.1	11.9	63.4	23.4	24.7 / 34.2 / 4.5	1
Qwen	2.3	10.5	50.9	16.3	27.7 / 21.9 / 1.3	0
Meta	2.4	6.5	77.5	25.6	6.7 / 70.4 / 0.3	0
Gemini	2.5	5.1	81.7	24.8	4.4 / 76.3 / 1.0	67
Gemini Smart	4.6	5.3	71.2	23.0	22.4 / 45.4 / 3.4	12
Deepgram	4.3	10.1	94.9	50.4	1.3 / 93.5 / 0.1	0
AAI 3.6	2.3	4.8	59.6	13.2	12.1 / 45.5 / 2.0	0
AAI 3.5	2.2	5.0	52.5	12.4	20.9 / 27.2 / 4.3	0
Zoom	2.3	7.7	38.7	10.3	12.8 / 24.3 / 1.5	0
ElevenLabs	2.8	4.6	66.6	19.4	33.9 / 25.7 / 7.0	0
ElevenLabs NV	2.7	4.6	66.9	18.8	32.4 / 27.2 / 7.3	0
GPT-Live	3.2	8.0	85.1	29.0	6.6 / 78.0 / 0.5	0
RT Whisper	3.0	8.8	93.6	41.2	2.0 / 91.6 / 0.0	0

Table 23: VOiCES: 100 sentences from 100 speakers, each as the clean original and as recorded by a far microphone in one room, with no background sound, with babble and with TV. WER (%), numbers and spellings normalised. Babble, no text: of the 100 babble clips, those with no transcript within 30 seconds, counted as empty.

G Two-Language Recordings: Details

Recordings each part of which every system handles alone. A part may go missing because it is hard on its own rather than because of the switch. Table 24 therefore repeats Table 9 on the recordings whose parts every one of the eleven systems gets at least 50% right

when played alone: 85 EN + X and 232 X + Y. Because this screen uses the tested systems themselves, it keeps recordings that are easy for all of them. Every rate is the same or up to 10 points lower, except ElevenLabs NV with automatic detection on EN + X (79% on all recordings, 80% on the screened ones). The order of the systems is the same, except that Meta and AAI 3.6 swap on X + Y with both languages selected (32% and 28% on all recordings, 22% and 24% on the screened ones). Adding Qwen to the screen removed no recording.

	EN + X (85 recordings)			X + Y (232 recordings)	
System	Both selected	English only	Auto-detect	Both selected	Auto-detect
Whisper SH	39	85	42	87	87
Qwen	14	33	14	43	43
Meta	13	16	15	22	25
Gemini	42	92	66	27	69
Gemini Smart	46	85	68	55	80
Deepgram	32	99	32	53	53
AAI 3.6	4	19	8	24	21
AAI 3.5	6	29	11	19	25
Zoom	56	56	56	81	81
ElevenLabs	61	84	61	92	91
ElevenLabs NV	80	93	80	96	96
ElevenLabs, pause	**1**	36	**1**	36	36
ElevenLabs NV, pause	2	46	2	62	63
GPT-Live	**1**	**1**	**1**	**0**	**0**
RT Whisper	5	5	4	4	5

Table 24: As Table 9, on the 85 EN + X and 232 X + Y recordings kept by the screen; lowest in each column in bold. The *pause* rows are ElevenLabs and ElevenLabs NV set to finalise at each pause of 300 ms, run on 29 October on these recordings only.

ElevenLabs and finalising. By default, ElevenLabs finalises each recording once, at the end, and loses the non-English part even of Spanish recordings that it transcribes when played alone, rarely writing anything in its place. We ran the test again on 29 October, on the screened recordings, with ElevenLabs set to finalise at each pause of 300 ms. This matches the 300 ms gap we put between the two clips, so switches without such a pause may behave differently, and the two runs are about two weeks apart. With both languages selected, recordings with a missing language fell from 61% to 1% with English (1 of 85; 95% CI 0 to 7%) and from 92% to 36% without (95% CI 26 to 47%); for ElevenLabs NV, from 80% to 2% and from 96% to 62%. Without English, the part still lost is almost always the second (80 of 83 recordings with both languages selected).

With both languages selected, systems that accept several languages are given both. Deepgram uses its multi-language mode for both settings; its documentation lists that mode for English, Spanish, French, German, Hindi, Russian, Portuguese, Japanese, Italian and Dutch [10], so Arabic and Mandarin are outside it. RT Whisper is given no language in either setting, and Zoom, which takes one language, is given none and uses its default, which behaves like English. So for these three, the both-languages and automatic settings send the same request. Deepgram's and Zoom's gave the same result on every recording; RT Whisper's differed on 1 of the 85 recordings with English and 2 of the 232 without. Qwen is sent both languages as `language_options` with both selected, and no language with automatic detection, but it wrote the same text in the two settings on all 400 recordings.

Setting	System	Non-English missing	First missing	Second missing	Nothing	Other text	Romanised	Translated
Both selected	Whisper SH	33	25	20	43	2	0	0
	Qwen	20	6	14	6	14	8	8
	Meta	15	0	19	18	1	0	0
	Gemini	38	35	16	45	6	10	0
	Gemini Smart	46	23	29	49	3	6	0
	Deepgram	40	17	23	18	22	34	4
	AAI 3.6	9	3	7	5	5	4	0
	AAI 3.5	12	5	8	7	6	2	2
	Zoom	62	30	32	55	7	6	0
	ElevenLabs	60	25	37	59	3	0	2
	ElevenLabs NV	77	34	46	73	7	2	2
	GPT-Live	6	0	6	4	2	2	0
	RT Whisper	10	2	9	8	3	0	0
English only	Whisper SH	86	44	43	64	23	4	18
	Qwen	38	20	18	17	21	10	14
	Meta	20	0	23	22	1	0	0
	Gemini	90	50	43	92	1	0	0
	Gemini Smart	86	36	51	85	2	2	2
	Deepgram	99	50	49	94	5	4	0
	AAI 3.6	25	10	15	18	7	6	2
	AAI 3.5	35	13	22	31	4	4	0
	Zoom	62	30	32	55	7	6	0
	ElevenLabs	85	50	35	82	3	2	2
	ElevenLabs NV	93	50	43	86	7	2	4
	GPT-Live	7	0	7	5	2	2	0
	RT Whisper	12	3	9	9	3	0	0
Auto-detect	Whisper SH	37	27	22	43	6	2	4
	Qwen	20	6	14	6	14	8	8
	Meta	17	0	21	20	1	0	0
	Gemini	36	54	19	70	3	4	0
	Gemini Smart	59	28	45	72	1	0	0
	Deepgram	40	17	23	18	22	34	4
	AAI 3.6	13	3	12	10	5	2	0
	AAI 3.5	13	5	11	11	5	0	2
	Zoom	62	30	32	55	7	6	0
	ElevenLabs	60	25	37	59	3	0	2
	ElevenLabs NV	76	33	46	73	6	2	2
	GPT-Live	7	1	6	6	1	0	0
	RT Whisper	10	2	8	6	4	0	0

Table 25: Missing parts in EN + X recordings, as a share (%) of all 100 recordings. Non-English missing: the part in the other language is missing (the measure of the filter and preference rule in Section 8). Nothing: no text where the part should be. Other text: different text in its place. Romanised and Translated are measured only on the 50 recordings whose other language has its own writing system (Arabic, Hindi or Mandarin), as a share of those 50. Romanised: the part is replaced by Latin-letter text that is not English. Translated: replaced by English words.

H Running Whisper: Further Results

Cutting at pauses, in full. Turning the gated deployment's VAD off is not enough: that deployment still has a whispered WER of 61.1%, almost all of it missing words (50% of words), and in 44% of whispered clips it drops 5 or more words from the last third. The self-hosted deployment, which transcribes exactly the segments the client cuts, loses almost nothing on the same clips (1.8% missing words, no cut-off endings). The two are different deployments, and the gated deployment's server still runs its own speech detection, so this comparison does not isolate the cutting rule from that second detector; we have not measured where that server cuts its segments or what it drops. On EARS the pattern repeats: with its VAD off, the gated deployment still loses 30.5% of whispered words, against 10.4% for the self-hosted deployment (Table 27). Cutting at the speaker's own pauses is also as fast as the gated setup: 361 ms typical wait against 360 ms, with a shorter slow case (498 ms against 596 ms).

Invented text, in full. Because of the two stock phrases (Section 9, Finding 4), the self-hosted deployment's WER on Spanish (Spain) is 19.5%, about twice the gated setup's 10.0%.

A VAD that decides only whether to send a segment at all prevents text on silence: Whisper-1 split wrote nothing on any no-speech clip. But the same rule cost it the whispered clips the VAD did not hear (23.4% against 7.7% for Whisper-1), so a VAD should not be allowed to skip segments that may contain quiet speech.

The nine setups in full.

1. **Gated, VAD on ("gated Whisper").** A deployment on Baseten with a voice gate on the client: a Silero VAD in front of Whisper drops audio it judges to be silence: it waits for speech above a 0.5 probability, keeps 96 ms before it, sends at most 512 ms of any pause, and stops sending after 544 ms of silence. The deployment cuts the audio into segments itself, with its own server-side speech detection, which we do not control.
2. **Gated, VAD off.** The same deployment, sent all of the audio: the client's Silero VAD is off, but the server's own speech detection still runs.
3. **Self-hosted, Silero cuts ("Whisper SH").** A second deployment of our own, which does no segmentation. The client runs the same VAD only to decide where each segment ends: at a pause of 544 ms after speech, once at least 5 seconds have been sent, and always before 30 seconds. It then tells the server to transcribe that segment. Every sample of audio is sent.
4. **Self-hosted, Silero drops.** The same deployment, with Silero VAD also dropping audio as in setup 1.
5. **Self-hosted, TEN cuts.** Setup 3 with TEN VAD [53], an open VAD from the TEN framework, in place of Silero, at its default threshold of 0.5. Every sample is sent.
6. **Self-hosted, TEN drops.** Setup 4 with TEN VAD: the same rules decide which audio is dropped, using TEN's speech probabilities instead of Silero's.
7. **OpenAI GPT-Realtime-Whisper ("RT Whisper").** OpenAI's hosted streaming Whisper.
8. **OpenAI Whisper-1.** OpenAI's file transcription, so the whole recording is uploaded when the user stops.
9. **Whisper-1 split.** Whisper-1 with the client cutting the recording at pauses, using the VAD rule of setup 3, and uploading each segment as soon as it is cut, with the previous segment's text as context. A segment in which the VAD heard no speech is not sent.

	Set	VAD on	VAD off	Change	95% CI
Gated Whisper	Quiet	8.7	9.5	+0.9	[+0.1, +1.6]
	Fan on	7.0	7.5	+0.6	[−1.3, +2.5]
	Whispered	80.7	61.1	−19.6	[−34.5, −7.5]
	Arabic (Emirati)	62.9	58.2	−4.7	[−10.6, −0.6]
	Whispered clips never sent	12.5	0.0		

Table 26: Gated Whisper with the client's Silero VAD on and off (the server's own speech detection runs in both), WER (%). Change: VAD off minus VAD on, in percentage points, with its 95% confidence interval from a paired comparison. Whispered clips never sent: share of the 16 whispered clips the VAD never sent.

The same setups across 107 speakers. Because our whispered set is 16 clips from one speaker, Table 27 repeats the setups that drop or cut audio on EARS (Section 6.2). On normal speech every setup is between 1.0% and 1.8%; on whispered speech they separate as on our own clips.

Setup	Normal	Whispered	Change (paired)	Whispered: missing	Clips never sent	Speakers ≥20%
Gated, VAD on	1.0	42.9 [36.0, 49.7]	+31.4 [+25.3, +37.5]	39.1	38	67
Gated, VAD off	1.2	32.8 [26.2, 39.1]	+21.3 [+15.6, +27.6]	30.5	0	53
Self-hosted, Silero cuts	1.5	11.5 [8.2, 15.2]	–	10.4	0	20
Self-hosted, Silero drops	1.5	38.3 [31.8, 45.0]	+26.8 [+21.3, +32.2]	36.1	0	62
Self-hosted, TEN cuts	1.3	9.5 [6.5, 13.1]	−2.0 [−3.7, −0.3]	8.6	0	17
Self-hosted, TEN drops	1.8	14.4 [10.1, 19.0]	+2.9 [−0.4, +6.7]	13.3	0	24

Table 27: Whisper setups on EARS: 107 speakers, 428 normal and 428 whispered clips. WER (%), numbers and spellings normalised as in Table 5, with 95% intervals from resampling speakers. Change: whispered WER minus that of the self-hosted deployment with Silero only cutting, paired by clip. Whispered, missing: missing words as a share of all words. Clips never sent: whispered clips the VAD never sent. Speakers ≥20%: speakers whose whispered WER is 20% or more.

TEN VAD, further results. On EARS, TEN also gives slightly lower WER than Silero when only cutting (9.5% against 11.5%, 95% CI of the difference −3.7 to −0.3). The reason is the one given in Section 9: TEN calls far more of the whispered audio speech than Silero does. On the no-speech clips, both call almost nothing speech at 0.5 (0.4% and 0%). Dropping with TEN also gives the self-hosted deployment a quiet-English WER of 7.2% (against 8.3% when sending all of the audio, within its interval) and an other-language average of 17.6%, lowers Spanish (Spain) from 19.5% to 14.7% (one clip that came back as "Gracias, señor presidente." is now transcribed correctly, though the other stock phrases remain), and reduces the no-speech clips with text from 6 to 2. The cost is a slightly longer wait (395 ms against 361 ms typical) and more lost languages in two-language recordings (54% against 45% with English, 96% against 87% without, with both languages selected), as with Silero dropping.

The threshold sweep. To check whether the Silero result is only a matter of its threshold, we repeated both dropping setups with the drop threshold at 0.2, 0.3 and 0.4, on the whispered, fan-on and no-speech clips (Figure 2). Lowering Silero's threshold helps only a little: whispered WER falls from 51.4% at 0.5 to 41.1% at 0.2, still four times the 10.1% of sending all of the audio, because even at 0.2 Silero calls only 52% of the whispered audio speech. TEN keeps whispered WER between 10.0% and 10.7% at every threshold. Neither VAD's fan-on WER moves by more than 2.1 points, and the no-speech clips with text stay at 0 of 10 for Silero and 2 of 10 for TEN at every threshold.

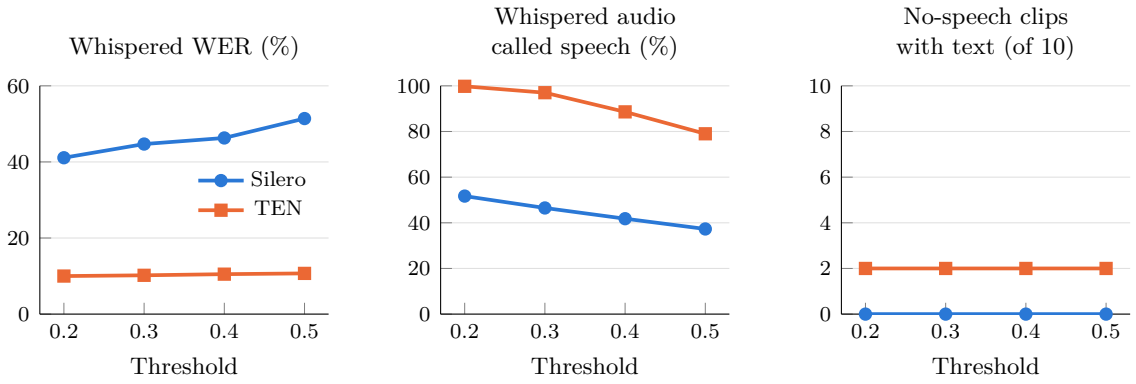


Figure 2: Self-hosted Whisper with the VAD dropping audio, at four drop thresholds. Left: whispered WER (sending all of the audio gives 10.1%). Middle: share of the whispered audio's 32 ms windows the VAD calls speech. Right: no-speech clips on which any text was written. One run per threshold.

I Further Limitations

The spoken references were drafted from the tested systems' own transcripts and confirmed by one listener, a member of the team, so a departure that no system heard, or that the listener missed, stays as in the script. Most analyses use the script, which favours the clean-up settings on extra words. Accents were labelled by a classifier trained on natural accents, not on accents a speaker puts on; we checked its labels only against the reader's own account. Names were labelled by a language model (Claude) following the written rule of Section 6.3, not by a person, and a listed word counts as a name wherever it appears in a message, so 5 of the 183 are lower-case uses of a product name. The spoken reference removes fillers ("um", "uh"), as the scripts do, so it is a record of the words said, not a full verbatim transcript. Romanised and translated parts are detected by rule, only when the two languages use different writing systems, and were checked by hand only on a few examples. The two-language recordings switch speaker and recording source along with the language, and the screened recordings of Appendix G were chosen using the tested systems themselves. The analysis thresholds (Table 1) are rules of thumb; other thresholds would change the counts.

Some choices were made on the test data: the two Whisper setups in the main comparison were chosen after the comparison in Section 9. GPT-Live's `delay` setting was first chosen on 50 of the English test clips; Table 14 now reports both settings on every test set.

Some systems and settings were not fully tested. Gated Whisper's two-language runs used its VAD. ElevenLabs was tested with its default settings and with `no_verbatim`; finalising at each pause was tested only on the two-language recordings. Other settings may change the results, as the Gemini Smart and ElevenLabs NV rows show. Zoom used the settings of its sample app. Whisper SH's segment cutting, and its use of TEN VAD, were implemented in our test client. The threshold sweep covered only the dropping setups, on the whispered, fan-on and no-speech clips, with one run per threshold, and only with the self-hosted deployment. Systems were tested on different days (Appendix B).

Each clip was played once per system and setting, except where stated, so we do not measure how much results vary between runs, apart from the one-week check in Appendix B and the repeated Gemini tests. All waits were measured from one client location, and the network round trip to each service differs (Table 22), so the waits are indicative rather than a ranking; repeated 9 to 28 days later, the median waits moved by up to 320 ms. Results for hosted services reflect their versions at the time of testing; the released transcripts record what each system returned then, but a rerun today may differ.